Set 1B: DC + AC Superposition

ECEN 325 — Lecture 1 (Linear Network Analysis) practice set

What this set trains. Circuits with DC supplies *and* an AC input. You split the problem in two (superposition): a DC circuit that gives $V_{o,dc}$ (capacitors open, inductors shorted) and an AC circuit that gives $H(s)$ and $v_{o,ac}(t)$ (supplies grounded). The DC part uses only one tool, the divider between two voltages. Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The Lecture-1 pipeline (use it for every problem in this set)

Tools — this is all the theory you need:

T1 *Impedances:* $Z_R = R$, $Z_C = \frac{1}{sC}$, $Z_L = sL$. Series: add. Parallel: $Z_a \parallel Z_b = \frac{Z_a Z_b}{Z_a + Z_b}$.

T2 *Voltage divider:* if Z_{top} connects V_{in} to a node and Z_{bot} is everything from that node to ground, then $V_{node} = V_{in}\,\frac{Z_{bot}}{Z_{top}+Z_{bot}}$.

T3 *Divider between two voltages (DC):* a node tied to V_a through R_a and to V_b through R_b sits at $V = V_b + (V_a - V_b)\frac{R_b}{R_a+R_b}$, and looks like a source V with resistance $R_a \parallel R_b$ (its Thevenin pair).

T4 *Dependent sources:* a VCVS forces $V = kV_x$; a VCCS drives the current gV_x through the impedance Z hanging on its node, so $V = \pm gV_xZ$ (sign from the arrow).

T5 *Sinusoids:* $A\sin(\omega t + \theta) \;\rightarrow\; |H(j\omega)|\,A\sin\big(\omega t + \theta + \angle H(j\omega)\big)$.

T6 *Bode rules:* a pole at p bends the magnitude by $-20\,\text{dB/dec}$ above p, a zero by $+20\,\text{dB/dec}$; $\frac{1}{1+s/p}$ is $0\,\text{dB}$ below p; $\frac{s}{s+p}$ is $0\,\text{dB}$ above p.

Steps:

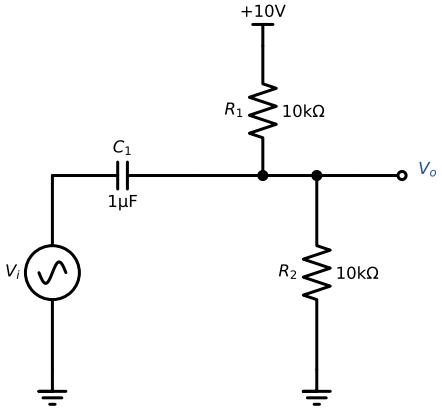
1. *Sources.* DC and AC both present $\Rightarrow$ superposition $V_o = V_{o,dc} + v_{o,ac}$.
2. *DC circuit:* $V_i \to 0$, $C \to$ open, $L \to$ short; reduce toward V_o with T3.
3. *AC circuit:* DC supplies $\to 0$ (rails become ground); merge parallel resistors.
4. *Divider chain:* name the nodes from V_i to V_o; $H = \prod$ (node-to-node ratios), each by T2 or T4, using the impedance seen looking right from each node.
5. *Standard form:* simplify, find the roots, write $H = K \cdot \prod(\text{factors})$.
6. *Bode:* start from the low-frequency slope, bend by $\pm 20\,\text{dB/dec}$ at each corner.
7. *Output:* evaluate each factor at $s = j\omega_{in}$, multiply magnitudes, add angles, apply T5 and add $V_{o,dc}$.

Problems

Easy problems

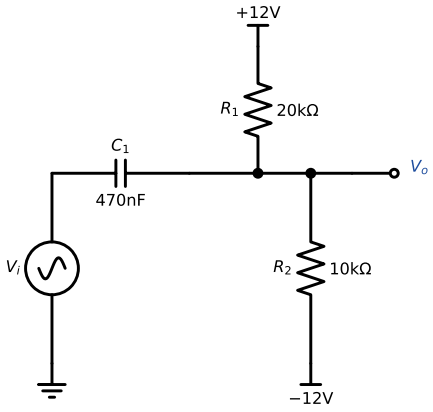
Problem 1B.1 — Easy

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5 \sin(2\pi \cdot 100\, t)$.



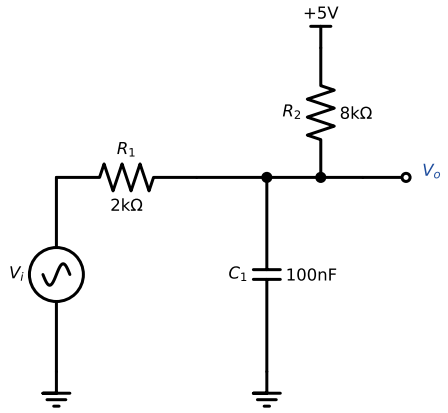
Problem 1B.2 — Easy

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1 \sin(2\pi \cdot 50\, t)$.



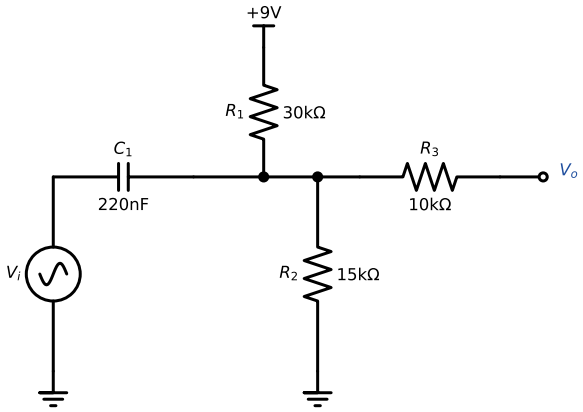
Problem 1B.3 — Easy

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 1000\,t)$.



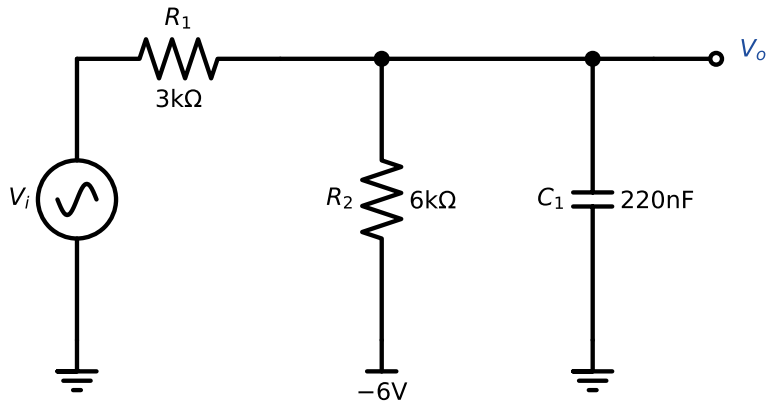
Problem 1B.4 — Easy

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin(2\pi \cdot 100\,t)$.



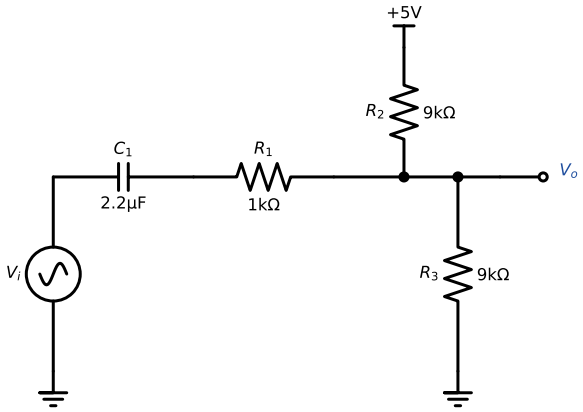
Problem 1B.5 — Easy

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 500\,t)$.



Problem 1B.6 — Easy

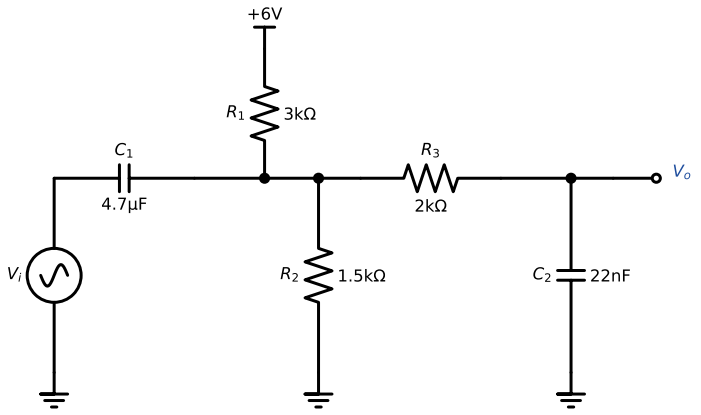
For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.3\sin(2\pi \cdot 20\,t)$.



Medium problems

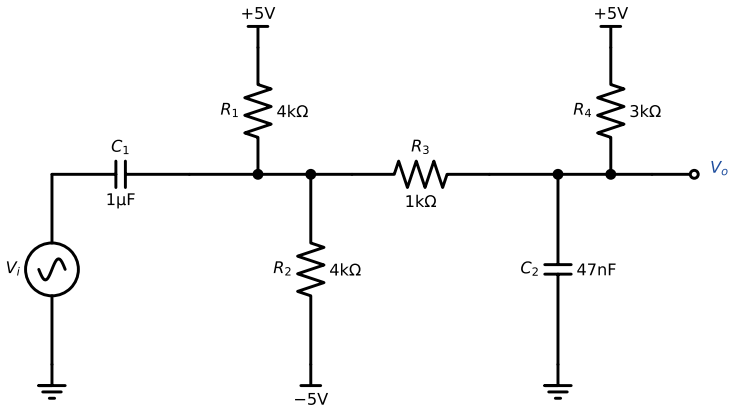
Problem 1B.7 — Medium

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 1000\,t)$.



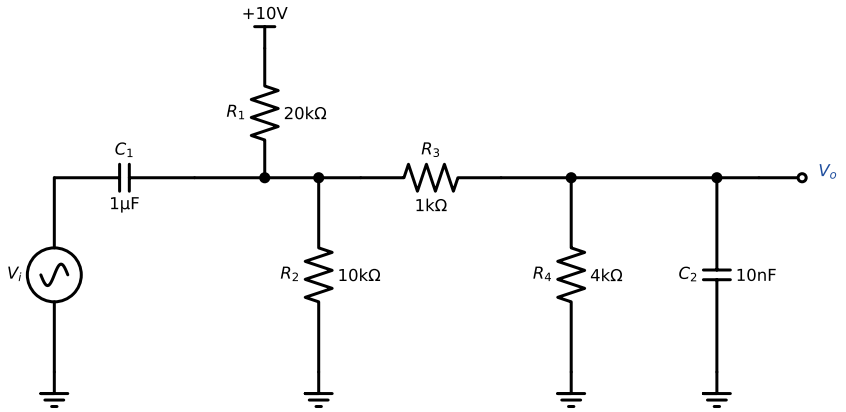
Problem 1B.8 — Medium

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 2000\,t)$.



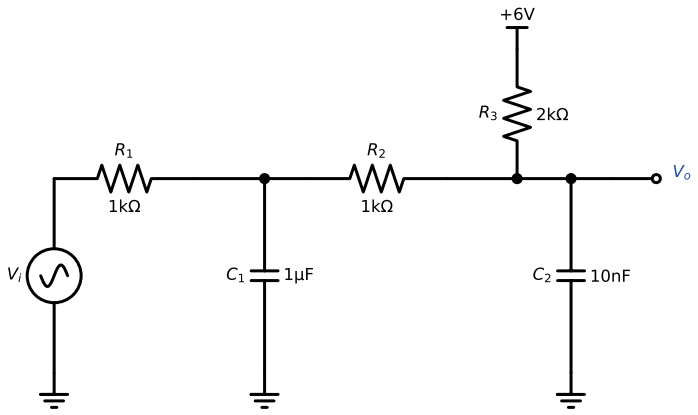
Problem 1B.9 — Medium

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 500\,t)$.



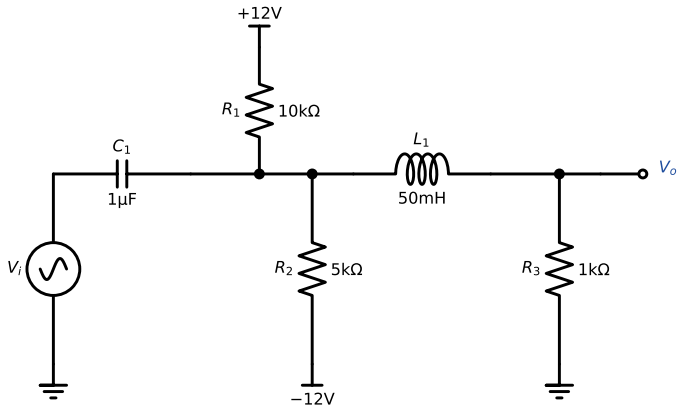
Problem 1B.10 — Medium

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 200\,t)$.



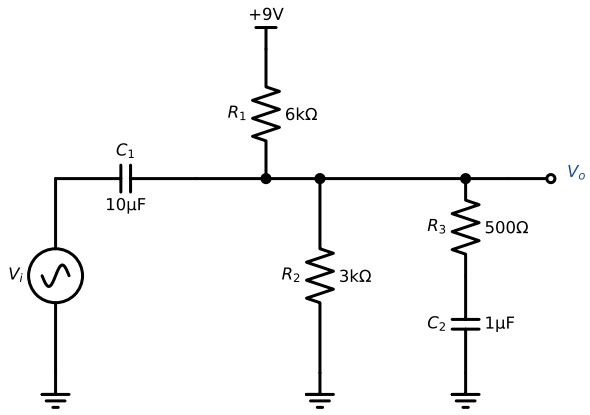
Problem 1B.11 — Medium

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 2\sin(2\pi \cdot 1000\,t)$.



Problem 1B.12 — Medium

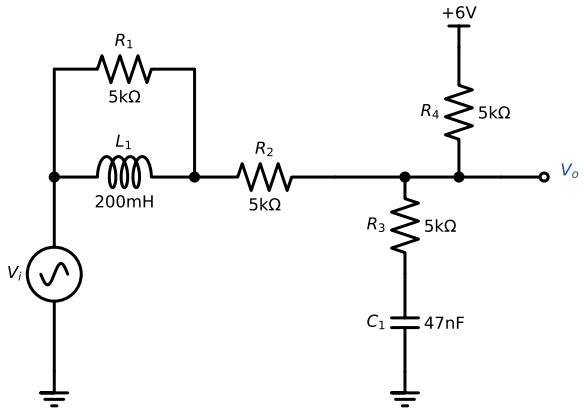
For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.4\sin(2\pi \cdot 300\,t)$.



Hard problems

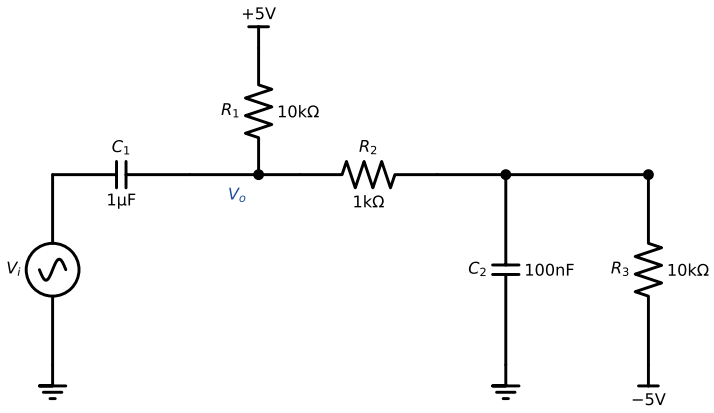
Problem 1B.13 — Hard

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1.5\sin\left(2\pi \cdot 1000\,t + \frac{\pi}{4}\right)$.



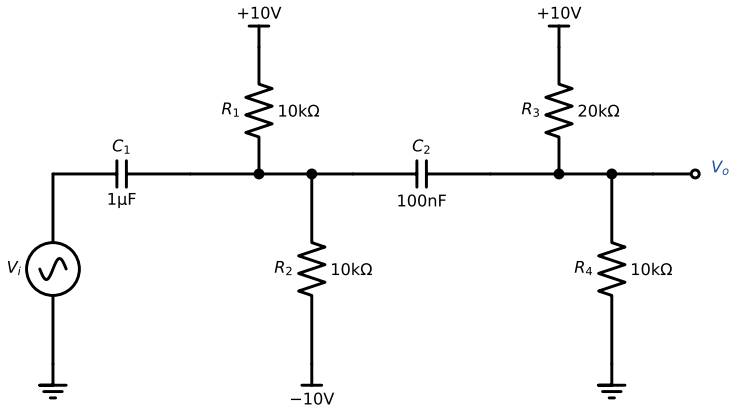
Problem 1B.14 — Hard

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin\left(2\pi \cdot 300\,t\right)$. Note that V_o is *not* at the right end: the branch to its right loads it, both at DC and at AC.



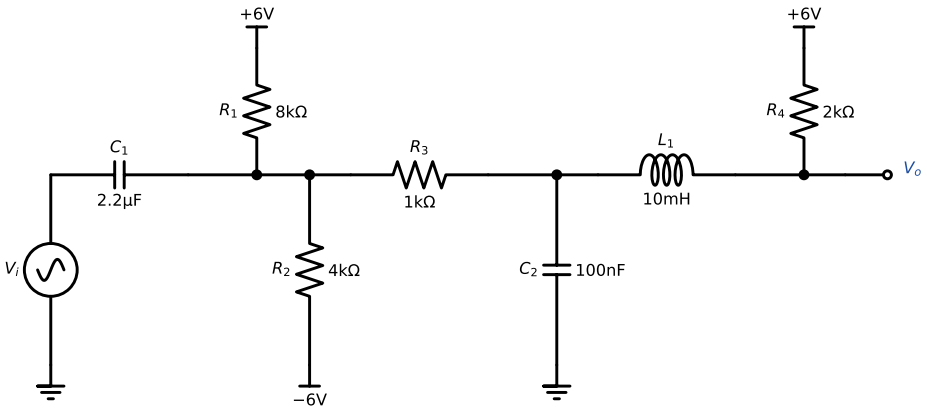
Problem 1B.15 — Hard

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.3\sin(2\pi \cdot 100\,t)$.



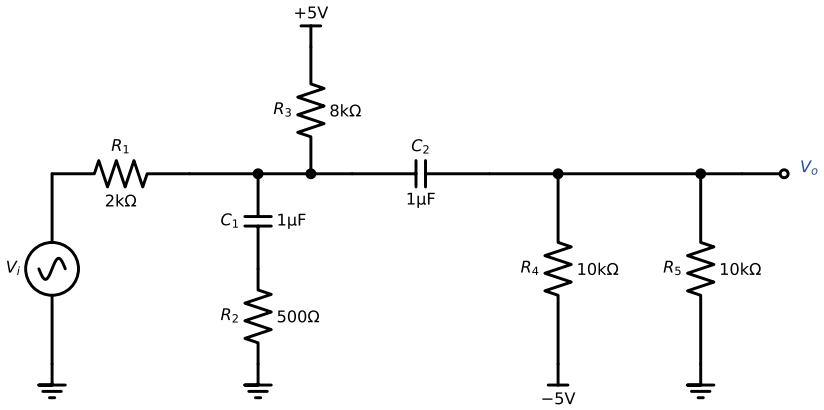
Problem 1B.16 — Hard

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\cos(2\pi \cdot 1000\,t)$.



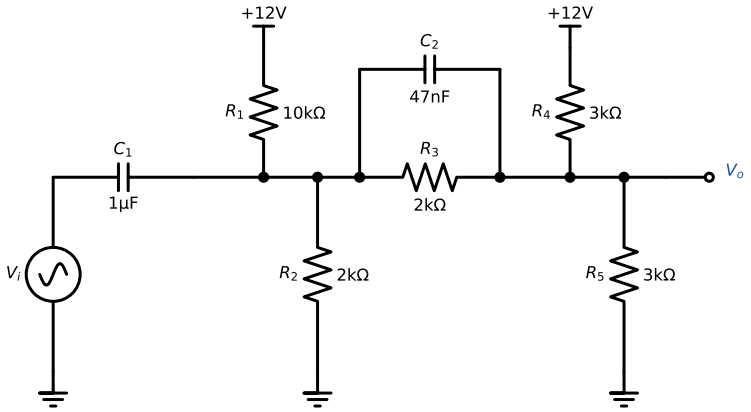
Problem 1B.17 — Hard

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\cos\left(2\pi \cdot 500\,t - \frac{\pi}{3}\right)$.



Problem 1B.18 — Hard

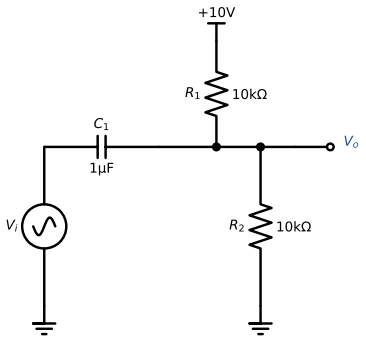
For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin\left(2\pi \cdot 2000\,t + \frac{\pi}{6}\right)$.



Solutions

Every solution follows the same seven steps of the pipeline card.

Solution 1B.1 **Easy**

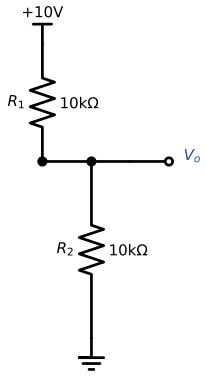


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies $+10\,\mathrm{V}$. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal $0\,\mathrm{V}$ source: $(V_{th}, R_{th}) = (0\,\mathrm{V},\, 0\,\Omega)$.
2. C_1 is a capacitor $\Rightarrow$ open at DC. No DC current can cross it, so everything on its left is disconnected: start fresh after it.
3. Finally, collect everything at the output node V_o.
4. First DC path here: R_1 to the $+10\,\mathrm{V}$ supply, so $(V_{th}, R_{th}) = (10\,\mathrm{V},\ 10\,\mathrm{k\Omega})$.
5. Add R_2 to ground. Divider between $V_a = 10\,\mathrm{V}$ (through $R_a = 10\,\mathrm{k\Omega}$) and $V_b = 0\,\mathrm{V}$ (through $R_b = 10\,\mathrm{k\Omega}$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 0 + (10 - (0))\,\frac{10\,\mathrm{k}}{10\,\mathrm{k} + 10\,\mathrm{k}} = 5\,\mathrm{V}, \qquad R_{th} = R_a \parallel R_b = 5\,\mathrm{k\Omega}$$

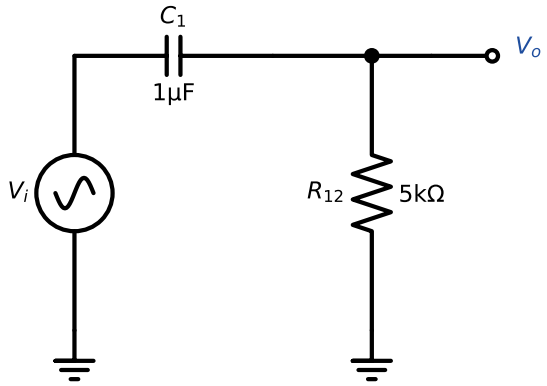
$$\boxed{V_{o,dc} = 5\,\mathrm{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1, R_2 both go from the same node to (AC) ground, so they are in parallel: $R_{12} = R_1 \parallel R_2 = 10\,\mathrm{k} \parallel 10\,\mathrm{k} = 5\,\mathrm{k\Omega}$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{R_{12}}{\frac{1}{sC_1} + R_{12}} = \frac{s}{s + 200}$$

So:

$$H(s) = \frac{s}{s + 200}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 200$; $s + 200 = 0 \Rightarrow s = -200$.

Factored form (leading coefficient in front):

$$H(s) = 1\,\frac{s}{(s + 200)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1\,\frac{s}{s+2\pi\cdot 31.8}, \qquad K = 1$$

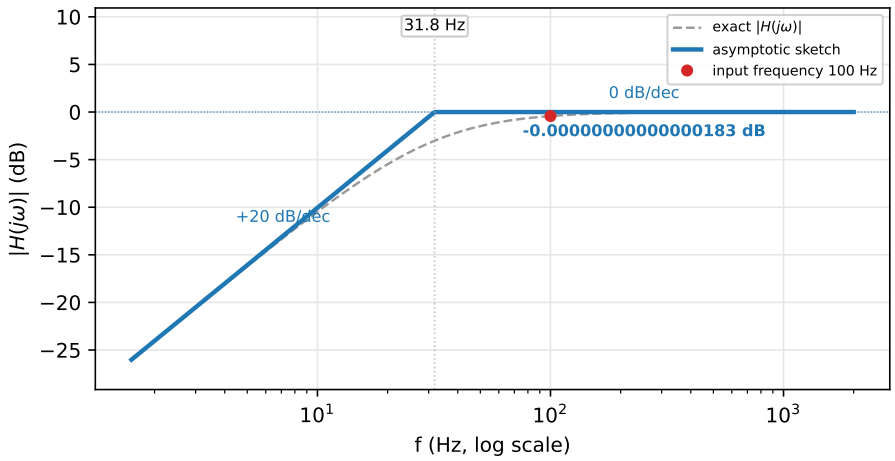
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	200	31.8 Hz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.005\,s$, so the plot starts with a slope of $+20\,\text{dB/dec}$. Then, moving right, each pole bends the line down by $20\,\text{dB/dec}$ and each zero bends it up by $20\,\text{dB/dec}$ (tool T6):

frequency range	slope	asymptote level
$f < 31.8$ Hz	$+20$ dB/dec	0 dB at end
$f > 31.8$ Hz	0 dB/dec (flat)	0 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1 = 0\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi\cdot 100\,t)$, so $\omega_{in} = 2\pi\cdot 100 = 628\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s+2\pi\cdot 31.8}$	0.9529	0.308
total	**0.9529** (product)	**0.308** (sum)

$$|H(j\omega_{in})| = 0.953, \qquad \angle H(j\omega_{in}) = 0.308 \text{ rad}$$
$$v_{o,ac}(t) = 0.953 \times 0.5 \sin(2\pi\cdot 100\,t + 0.308) = 0.476 \sin(2\pi\cdot 100\,t + 0.308)$$

Add the DC part from Step 2 (superposition):

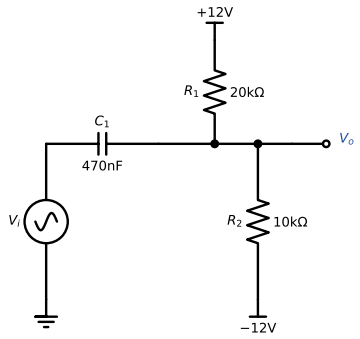
$$\boxed{V_o(t) = 5 + 0.476 \sin(2\pi\cdot 100\,t + 0.308) \text{ V}}$$

Answer

$H(s) = 1\,\dfrac{s}{s+2\pi\cdot 31.8}$ (corners: 31.8 Hz)

$V_o(t) = 5 + 0.476 \sin(2\pi\cdot 100\,t + 0.308)$ V

Solution 1B.2 **Easy**

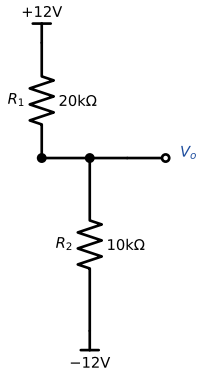


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies $-12\,\text{V}$, $+12\,\text{V}$. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal $0\,\text{V}$ source: $(V_{th}, R_{th}) = (0\,\text{V},\, 0\,\Omega)$.
2. C_1 is a capacitor $\Rightarrow$ open at DC. No DC current can cross it, so everything on its left is disconnected: start fresh after it.
3. Finally, collect everything at the output node V_o.
4. First DC path here: R_1 to the $+12\,\text{V}$ supply, so $(V_{th}, R_{th}) = (12\,\text{V},\ 20\,\text{k}\Omega)$.
5. Add R_2 to the $-12\,\text{V}$ supply. Divider between $V_a = 12\,\text{V}$ (through $R_a = 20\,\text{k}\Omega$) and $V_b = -12\,\text{V}$ (through $R_b = 10\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = -12 + (12 - (-12))\,\frac{10\,\text{k}}{20\,\text{k} + 10\,\text{k}} = -4\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 6.67\,\text{k}\Omega$$

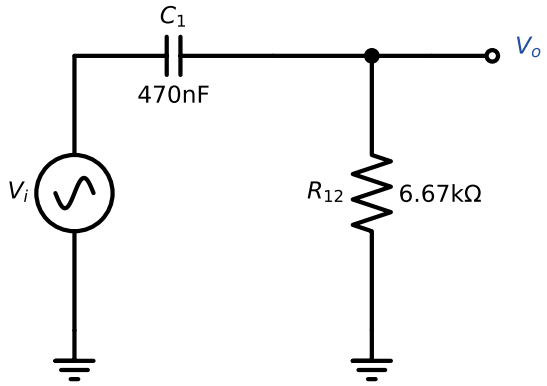
$$\boxed{V_{o,dc} = -4\,\mathrm{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1, R_2 both go from the same node to (AC) ground, so they are in parallel: $R_{12} = R_1 \parallel R_2 = 20\,\mathrm{k} \parallel 10\,\mathrm{k} = 6.67\,\mathrm{k\Omega}$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{R_{12}}{\frac{1}{sC_1} + R_{12}} = \frac{s}{s + 319}$$

So:

$$H(s) = \frac{s}{s + 319}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 319$; $s + 319 = 0 \Rightarrow s = -319$.

Factored form (leading coefficient in front):

$$H(s) = 1\,\frac{s}{(s + 319)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1\,\frac{s}{s + 2\pi \cdot 50.8}, \qquad K = 1$$

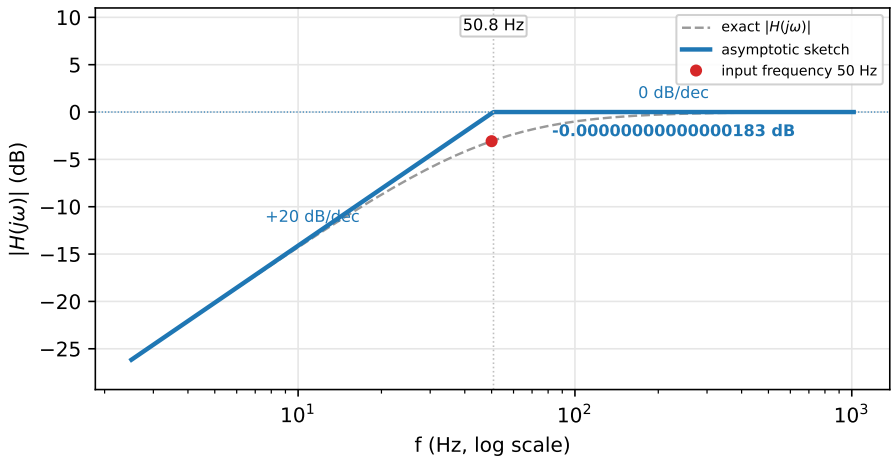
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	319	50.8 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.00313\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 50.8$ Hz	+20 dB/dec	0 dB at end
$f > 50.8$ Hz	0 dB/dec (flat)	0 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1 = 0$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi \cdot 50\,t)$, so $\omega_{in} = 2\pi \cdot 50 = 314$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s+2\pi\cdot 50.8}$	0.7015	0.793
total	**0.7015** (product)	**0.793** (sum)

$$|H(j\omega_{in})| = 0.702, \qquad \angle H(j\omega_{in}) = 0.793 \text{ rad}$$
$$v_{o,ac}(t) = 0.702 \times 1 \sin\left(2\pi\cdot 50\,t + 0.793\right) = 0.702 \sin\left(2\pi\cdot 50\,t + 0.793\right)$$

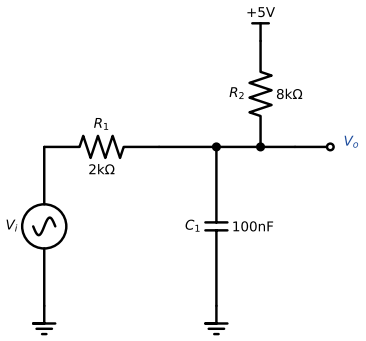
Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = -4 + 0.702 \sin\left(2\pi\cdot 50\,t + 0.793\right) \text{ V}}$$

Answer

$$H(s) = 1\,\frac{s}{s+2\pi\cdot 50.8} \quad \text{(corners: } 50.8 \text{ Hz)}$$
$$V_o(t) = -4 + 0.702 \sin\left(2\pi\cdot 50\,t + 0.793\right) \text{ V}$$

Solution 1B.3 **Easy**

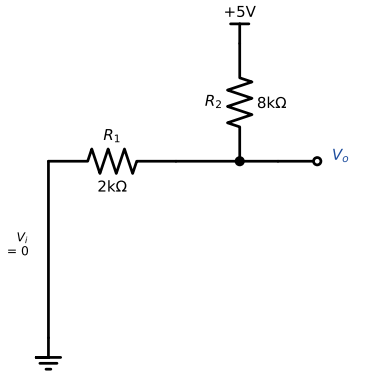


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies +5 V. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

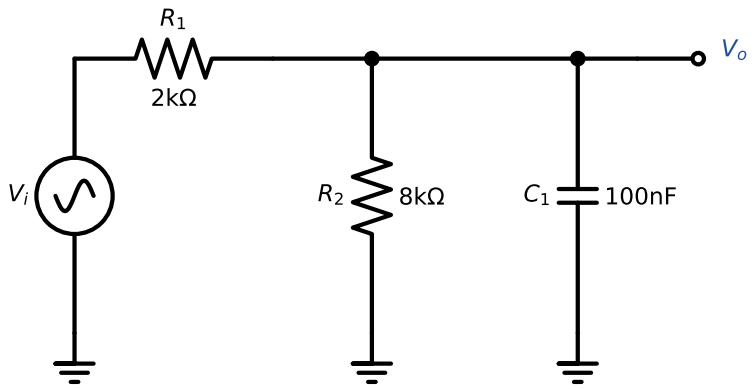
1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal 0 V source: $(V_{th}, R_{th}) = (0\,\text{V},\, 0\,\Omega)$.
2. Walk through R_1 (2 kΩ at DC). A series resistor just adds to R_{th}: $(V_{th}, R_{th}) = (0\,\text{V},\ 2\,\text{k}\Omega)$.
3. Finally, collect everything at the output node V_o.
4. C_1 (at V_o) is a capacitor: open at DC, ignore it.
5. Add R_2 to the +5 V supply. Divider between $V_a = 0\,\text{V}$ (through $R_a = 2\,\text{k}\Omega$) and $V_b = 5\,\text{V}$ (through $R_b = 8\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 5 + (0 - (5))\,\frac{8\,\text{k}}{2\,\text{k} + 8\,\text{k}} = 1\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 1.6\,\text{k}\Omega$$

$$\boxed{V_{o,dc} = 1\,\text{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_2 \parallel \frac{1}{sC_1} = \frac{10^7}{s + 1250}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{R_1 + Z_o} = \frac{5000}{s + 6250}$$

So:

$$H(s) = \frac{5000}{s + 6250}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 6250$; $s + 6250 = 0 \Rightarrow s = -6250$.

Factored form (leading coefficient in front):

$$H(s) = 5000\,\frac{1}{(s + 6250)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.8\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 995}}, \qquad K = 0.8$$

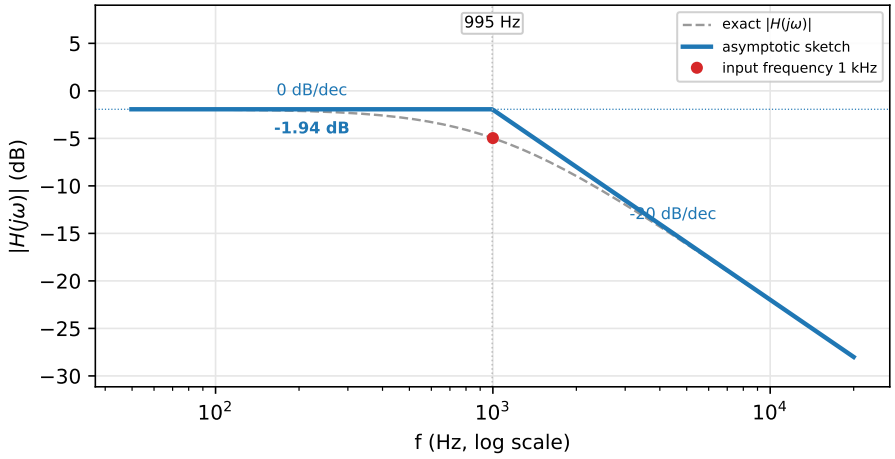
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	6250	995 Hz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 0.8 = -1.94\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 995$ Hz	0 dB/dec (flat)	-1.94 dB at end
$f > 995$ Hz	-20 dB/dec	-1.94 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}0.8 = -1.94\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.8	0.8	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 995}}$	0.7052	-0.788
total	**0.5642** (product)	$\mathbf{-0.788}$ (sum)

$$|H(j\omega_{in})| = 0.564, \qquad \angle H(j\omega_{in}) = -0.788 \text{ rad}$$

$$v_{o,ac}(t) = 0.564 \times 0.5 \sin\left(2\pi \cdot 1000\, t - 0.788\right) = 0.282 \sin\left(2\pi \cdot 1000\, t - 0.788\right)$$

Add the DC part from Step 2 (superposition):

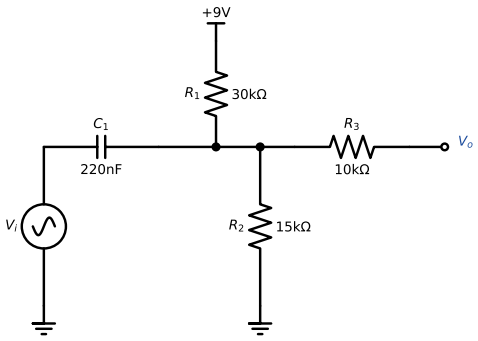
$$\boxed{V_o(t) = 1 + 0.282 \sin\left(2\pi \cdot 1000\, t - 0.788\right)\ \text{V}}$$

Answer

$$H(s) = 0.8\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 995}} \quad \text{(corners: 995 Hz)}$$

$$V_o(t) = 1 + 0.282 \sin\left(2\pi \cdot 1000\, t - 0.788\right)\ \text{V}$$

Solution 1B.4 **Easy**

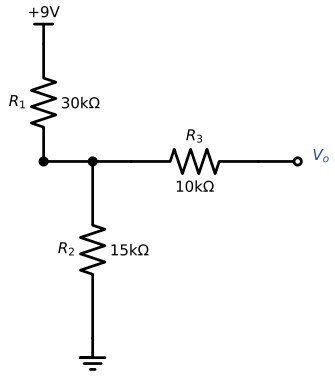


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies +9 V. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal 0 V source: $(V_{th}, R_{th}) = (0\,\text{V},\, 0\,\Omega)$.
2. C_1 is a capacitor $\Rightarrow$ open at DC. No DC current can cross it, so everything on its left is disconnected: start fresh after it.
3. First DC path here: R_1 to the +9 V supply, so $(V_{th}, R_{th}) = (9\,\text{V},\ 30\,\text{k}\Omega)$.
4. Add R_2 to ground. Divider between $V_a = 9$ V (through $R_a = 30\,\text{k}\Omega$) and $V_b = 0$ V (through $R_b = 15\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 0 + (9 - (0))\,\frac{15\,\text{k}}{30\,\text{k} + 15\,\text{k}} = 3\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 10\,\text{k}\Omega$$

5. Walk through R_3 (10 kΩ at DC). A series resistor just adds to R_{th}: $(V_{th}, R_{th}) = (3\,\text{V},\ 20\,\text{k}\Omega)$.

6. Finally, collect everything at the output node V_o.
7. Nothing else draws DC current from V_o, so no current flows through R_{th} and there is no voltage drop across it: $V_{o,dc} = V_{th}$.

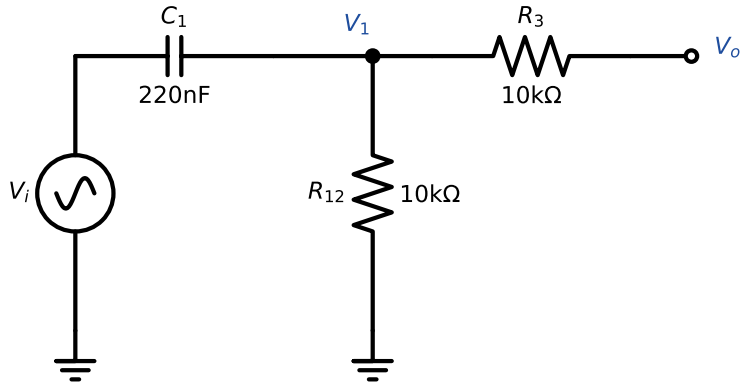
$$\boxed{V_{o,dc} = 3\,\mathrm{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1, R_2 both go from the same node to (AC) ground, so they are in parallel: $R_{12} = R_1 \parallel R_2 = 30\,\mathrm{k} \parallel 15\,\mathrm{k} = 10\,\mathrm{k\Omega}$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{R_{12}}{\frac{1}{sC_1} + R_{12}} = \frac{s}{s + 455}$$

$$\frac{V_o}{V_1} = 1 \qquad \text{(nothing connected after } R_3\text{: no current, no drop)}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{s}{s + 455}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 455$; $s + 455 = 0 \Rightarrow s = -455$.

Factored form (leading coefficient in front):

$$H(s) = 1\,\frac{s}{(s + 455)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1\,\frac{s}{s + 2\pi \cdot 72.3}, \qquad K = 1$$

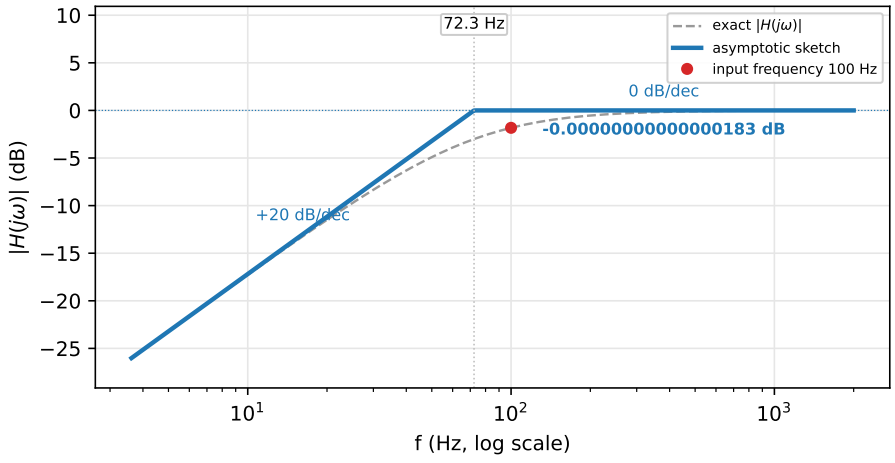
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	455	72.3 Hz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.0022\,s$, so the plot starts with a slope of $+20\,\text{dB/dec}$. Then, moving right, each pole bends the line down by $20\,\text{dB/dec}$ and each zero bends it up by $20\,\text{dB/dec}$ (tool T6):

frequency range	slope	asymptote level
$f < 72.3$ Hz	$+20$ dB/dec	0 dB at end
$f > 72.3$ Hz	0 dB/dec (flat)	0 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1 = 0\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin(2\pi \cdot 100\,t)$, so $\omega_{in} = 2\pi \cdot 100 = 628\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s+2\pi \cdot 72.3}$	0.8102	0.626
total	**0.8102** (product)	**0.626** (sum)

$$|H(j\omega_{in})| = 0.81, \qquad \angle H(j\omega_{in}) = 0.626 \text{ rad}$$
$$v_{o,ac}(t) = 0.81 \times 0.2 \sin(2\pi \cdot 100\,t + 0.626) = 0.162 \sin(2\pi \cdot 100\,t + 0.626)$$

Add the DC part from Step 2 (superposition):

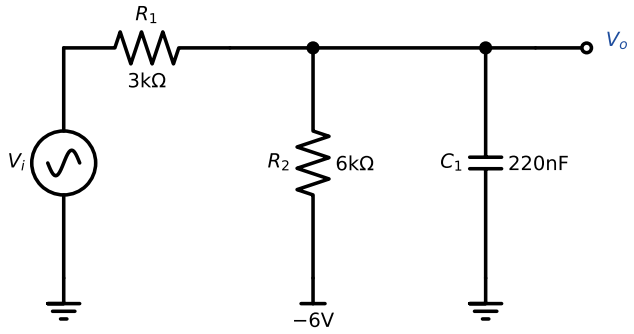
$$\boxed{V_o(t) = 3 + 0.162 \sin(2\pi \cdot 100\,t + 0.626)\ \text{V}}$$

Answer

$H(s) = 1\,\dfrac{s}{s+2\pi \cdot 72.3}$ (corners: 72.3 Hz)

$V_o(t) = 3 + 0.162 \sin(2\pi \cdot 100\,t + 0.626)$ V

Solution 1B.5 **Easy**

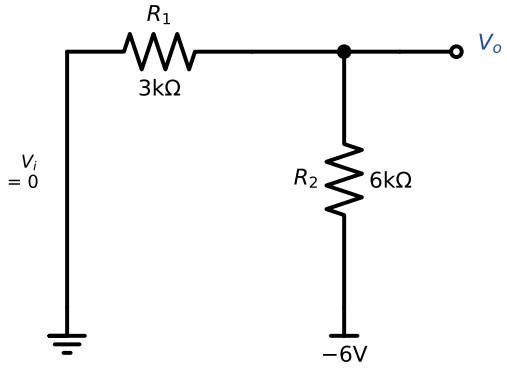


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies $-6\,\mathrm{V}$. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal $0\,\mathrm{V}$ source: $(V_{th}, R_{th}) = (0\,\mathrm{V},\, 0\,\Omega)$.
2. Walk through R_1 ($3\,\mathrm{k}\Omega$ at DC). A series resistor just adds to R_{th}: $(V_{th}, R_{th}) = (0\,\mathrm{V},\ 3\,\mathrm{k}\Omega)$.
3. Finally, collect everything at the output node V_o.
4. Add R_2 to the $-6\,\mathrm{V}$ supply. Divider between $V_a = 0\,\mathrm{V}$ (through $R_a = 3\,\mathrm{k}\Omega$) and $V_b = -6\,\mathrm{V}$ (through $R_b = 6\,\mathrm{k}\Omega$):

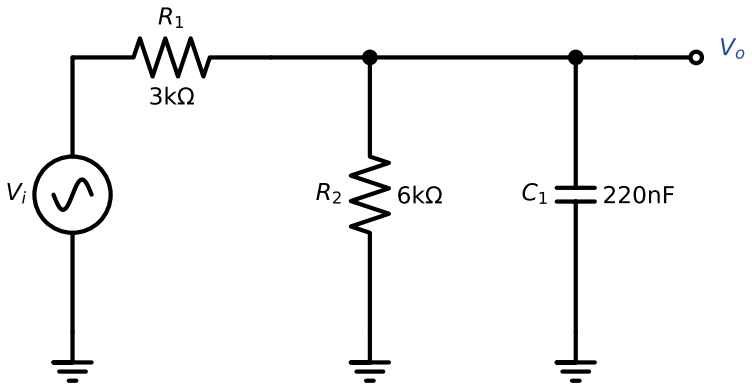
$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = -6 + (0 - (-6))\,\frac{6\,\mathrm{k}}{3\,\mathrm{k} + 6\,\mathrm{k}} = -2\,\mathrm{V}, \qquad R_{th} = R_a \parallel R_b = 2\,\mathrm{k}\Omega$$

5. C_1 (at V_o) is a capacitor: open at DC, ignore it.

$$\boxed{V_{o,dc} = -2\,\mathrm{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_2 \parallel \frac{1}{sC_1} = \frac{4.55 \times 10^6}{s + 758}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{R_1 + Z_o} = \frac{1520}{s + 2270}$$

So:

$$H(s) = \frac{1520}{s + 2270}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 2270$; $s + 2270 = 0 \Rightarrow s = -2270$.

Factored form (leading coefficient in front):

$$H(s) = 1520\,\frac{1}{(s + 2270)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.667\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 362}}, \qquad K = 0.667$$

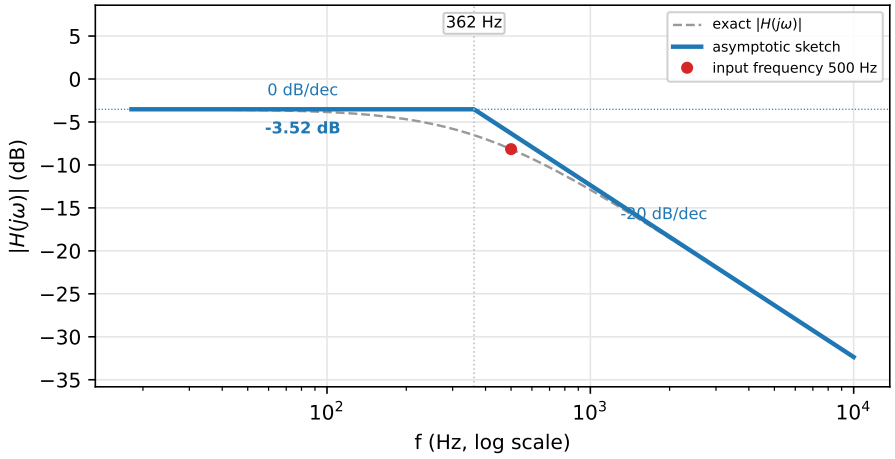
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	2270	362 Hz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 0.667 = -3.52\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 362$ Hz	0 dB/dec (flat)	-3.52 dB at end
$f > 362$ Hz	-20 dB/dec	-3.52 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 0.667 = -3.52\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi \cdot 500\,t)$, so $\omega_{in} = 2\pi \cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.667	0.6667	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 362}}$	0.5861	-0.945
total	**0.3908** (product)	$\mathbf{-0.945}$ (sum)

$$|H(j\omega_{in})| = 0.391, \qquad \angle H(j\omega_{in}) = -0.945 \text{ rad}$$

$$v_{o,ac}(t) = 0.391 \times 1 \sin\left(2\pi \cdot 500\, t - 0.945\right) = 0.391 \sin\left(2\pi \cdot 500\, t - 0.945\right)$$

Add the DC part from Step 2 (superposition):

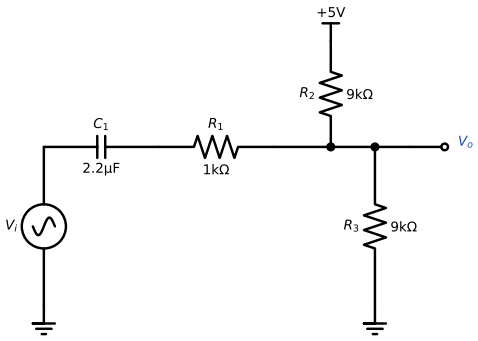
$$\boxed{V_o(t) = -2 + 0.391 \sin\left(2\pi \cdot 500\, t - 0.945\right) \text{ V}}$$

Answer

$$H(s) = 0.667 \frac{1}{1 + \dfrac{s}{2\pi \cdot 362}} \quad \text{(corners: 362 Hz)}$$

$$V_o(t) = -2 + 0.391 \sin\left(2\pi \cdot 500\, t - 0.945\right) \text{ V}$$

Solution 1B.6 **Easy**

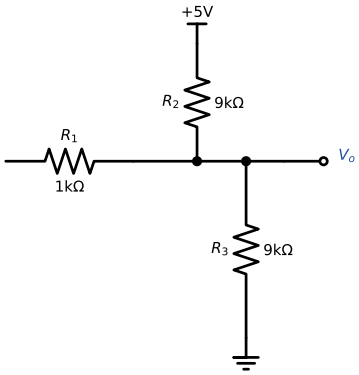


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies +5 V. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal 0 V source: $(V_{th}, R_{th}) = (0\,\text{V},\, 0\,\Omega)$.
2. C_1 is a capacitor $\Rightarrow$ open at DC. No DC current can cross it, so everything on its left is disconnected: start fresh after it.
3. Finally, collect everything at the output node V_o.
4. First DC path here: R_2 to the +5 V supply, so $(V_{th}, R_{th}) = (5\,\text{V},\ 9\,\text{k}\Omega)$.
5. Add R_3 to ground. Divider between $V_a = 5\,\text{V}$ (through $R_a = 9\,\text{k}\Omega$) and $V_b = 0\,\text{V}$ (through $R_b = 9\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 0 + (5 - (0))\,\frac{9\,\text{k}}{9\,\text{k} + 9\,\text{k}} = 2.5\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 4.5\,\text{k}\Omega$$

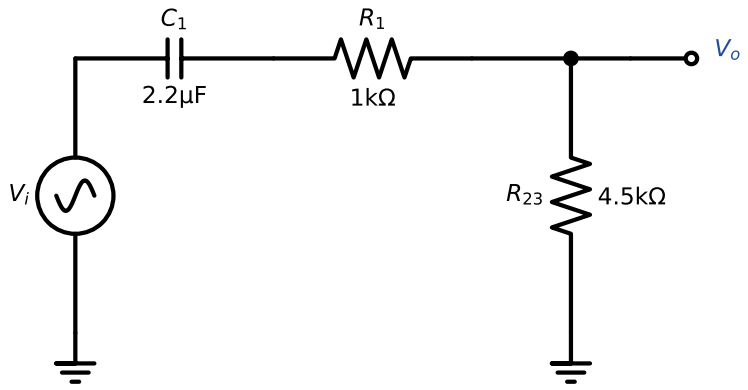
$$\boxed{V_{o,dc} = 2.5\,\text{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_2, R_3 both go from the same node to (AC) ground, so they are in parallel: $R_{23} = R_2 \parallel R_3 = 9\,\text{k} \parallel 9\,\text{k} = 4.5\,\text{k}\Omega$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{R_{23}}{\frac{1}{sC_1} + R_1 + R_{23}} = \frac{0.818\,s}{s + 82.6}$$

So:

$$H(s) = \frac{0.818\,s}{s + 82.6}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 82.6$; $s + 82.6 = 0 \Rightarrow s = -82.6$.

Factored form (leading coefficient in front):

$$H(s) = 0.818\,\frac{s}{(s + 82.6)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.818\,\frac{s}{s + 2\pi \cdot 13.2}, \qquad K = 0.818$$

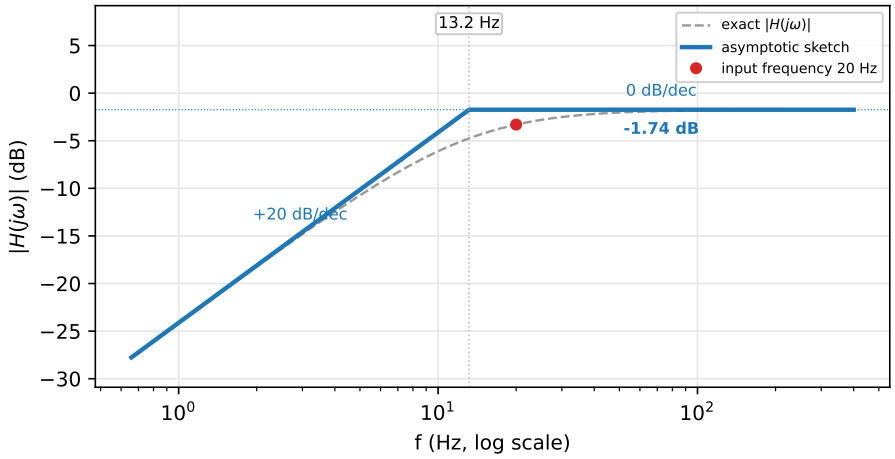
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	82.6	13.2 Hz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.0099\,s$, so the plot starts with a slope of $+20\,\text{dB/dec}$. Then, moving right, each pole bends the line down by $20\,\text{dB/dec}$ and each zero bends it up by $20\,\text{dB/dec}$ (tool T6):

frequency range	slope	asymptote level
$f < 13.2$ Hz	$+20$ dB/dec	-1.74 dB at end
$f > 13.2$ Hz	0 dB/dec (flat)	-1.74 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 0.818 = -1.74\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.3\sin(2\pi \cdot 20\,t)$, so $\omega_{in} = 2\pi \cdot 20 = 126\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.818	0.8182	0
$\dfrac{s}{s + 2\pi \cdot 13.2}$	0.8355	0.582
total	**0.6836** (product)	**0.582** (sum)

$$|H(j\omega_{in})| = 0.684, \qquad \angle H(j\omega_{in}) = 0.582 \text{ rad}$$

$$v_{o,ac}(t) = 0.684 \times 0.3 \sin(2\pi \cdot 20\,t + 0.582) = 0.205 \sin(2\pi \cdot 20\,t + 0.582)$$

Add the DC part from Step 2 (superposition):

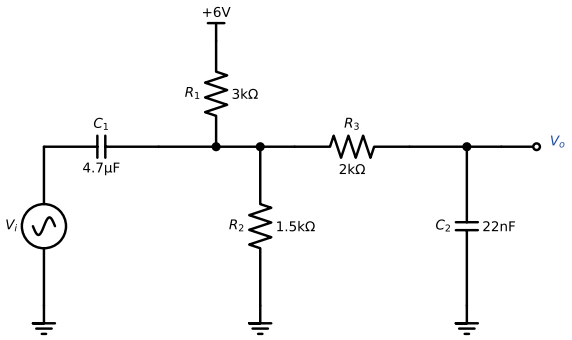
$$\boxed{V_o(t) = 2.5 + 0.205 \sin(2\pi \cdot 20\,t + 0.582) \ \text{V}}$$

Answer

$H(s) = 0.818 \dfrac{s}{s + 2\pi \cdot 13.2}$ (corners: 13.2 Hz)

$V_o(t) = 2.5 + 0.205 \sin(2\pi \cdot 20\,t + 0.582)$ V

Solution 1B.7 **Medium**

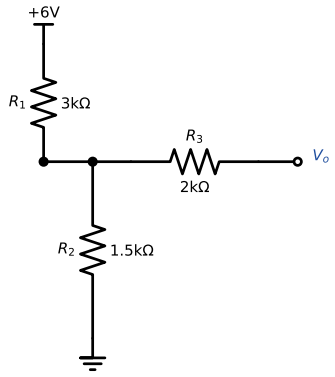


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies +6 V. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1, C_2). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal 0 V source: $(V_{th}, R_{th}) = (0\,\text{V},\, 0\,\Omega)$.
2. C_1 is a capacitor $\Rightarrow$ open at DC. No DC current can cross it, so everything on its left is disconnected: start fresh after it.
3. First DC path here: R_1 to the +6 V supply, so $(V_{th}, R_{th}) = (6\,\text{V},\ 3\,\text{k}\Omega)$.
4. Add R_2 to ground. Divider between $V_a = 6$ V (through $R_a = 3\,\text{k}\Omega$) and $V_b = 0$ V (through $R_b = 1.5\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 0 + (6 - (0))\,\frac{1.5\,\text{k}}{3\,\text{k} + 1.5\,\text{k}} = 2\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 1\,\text{k}\Omega$$

5. Walk through R_3 (2 kΩ at DC). A series resistor just adds to R_{th}: $(V_{th}, R_{th}) = (2\,\text{V},\ 3\,\text{k}\Omega)$.
6. Finally, collect everything at the output node V_o.

7. C_2 (at V_o) is a capacitor: open at DC, ignore it.
8. Nothing else draws DC current from V_o, so no current flows through R_{th} and there is no voltage drop across it: $V_{o,dc} = V_{th}$.

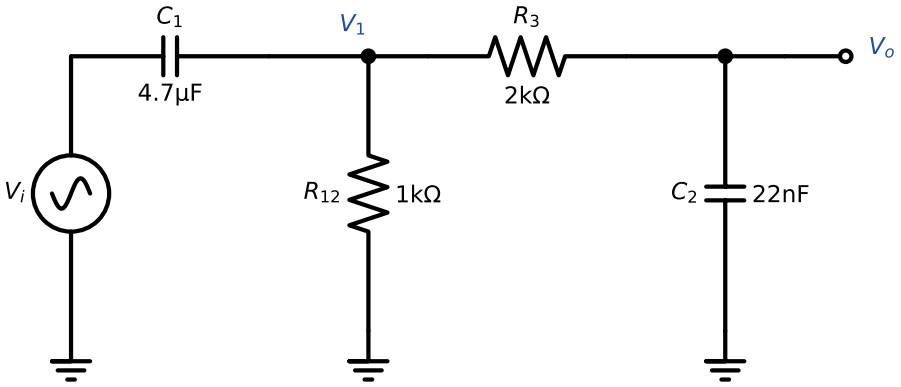
$$\boxed{V_{o,dc} = 2\,\mathrm{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1, R_2 both go from the same node to (AC) ground, so they are in parallel: $R_{12} = R_1 \parallel R_2 = 3\,\mathrm{k} \parallel 1.5\,\mathrm{k} = 1\,\mathrm{k\Omega}$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_{12} \parallel \left(R_3 + \frac{1}{sC_2}\right) = \frac{667\,s + 1.52 \times 10^7}{s + 15200}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^2 + 22700\,s}{s^2 + 23000\,s + 4.84 \times 10^6}$$

$$\frac{V_o}{V_1} = \frac{\frac{1}{sC_2}}{R_3 + \frac{1}{sC_2}} = \frac{22700}{s + 22700}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{22700\, s}{s^2 + 23000\, s + 4.84 \times 10^6}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 23000\, s + 4.84 \times 10^6$; quadratic formula:

$$s = \frac{-23000 \pm \sqrt{(23000)^2 - 4(4.84 \times 10^6)}}{2} = \frac{-23000 \pm 22600}{2} \Rightarrow s = -212,\ -22800$$

Factored form (leading coefficient in front):

$$H(s) = 22700\, \frac{s}{(s + 212)(s + 22800)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.995\, \frac{s}{s + 2\pi \cdot 33.7}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 3.63\text{k}}}, \qquad K = 0.995$$

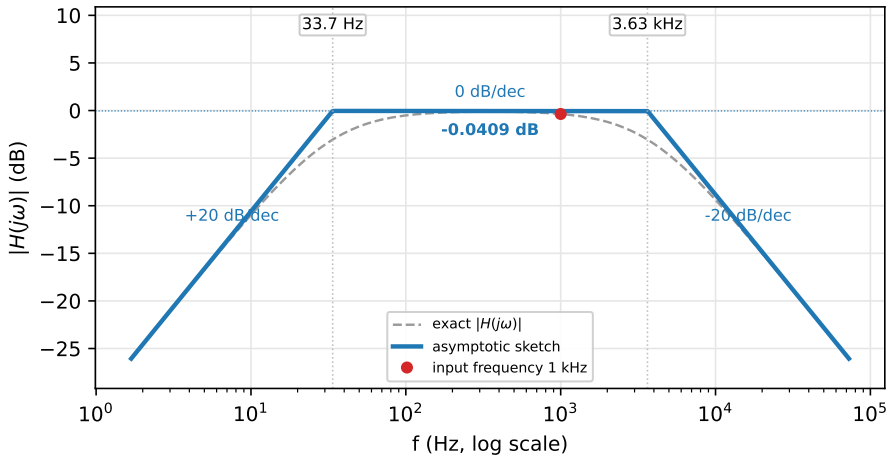
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	212	33.7 Hz	−20 dB/dec
pole	22800	3.63 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.0047\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 33.7 Hz	+20 dB/dec	-0.0409 dB at end
33.7 Hz – 3.63 kHz	0 dB/dec (flat)	-0.0409 dB at start, -0.0409 dB at end
$f >$ 3.63 kHz	−20 dB/dec	-0.0409 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 0.995 = -0.0409$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.995	0.9953	0
$\dfrac{s}{s+2\pi\cdot 33.7}$	0.9994	0.0337
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 3.63\text{k}}}$	0.9642	-0.269
total	**0.9591** (product)	**−0.235** (sum)

$$|H(j\omega_{in})| = 0.959, \qquad \angle H(j\omega_{in}) = -0.235 \text{ rad}$$
$$v_{o,ac}(t) = 0.959 \times 0.5\sin(2\pi\cdot 1000\,t - 0.235) = 0.48\sin(2\pi\cdot 1000\,t - 0.235)$$

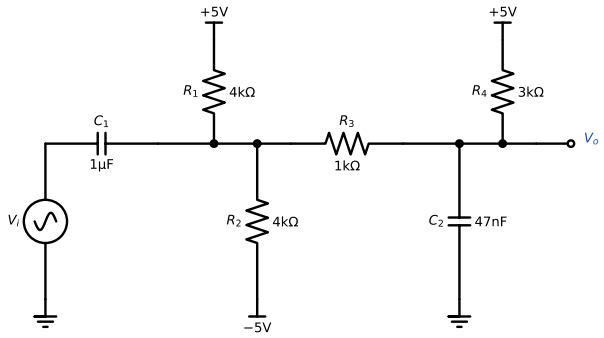
Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = 2 + 0.48\sin(2\pi\cdot 1000\,t - 0.235)\ \text{V}}$$

Answer

$$H(s) = 0.995\,\frac{s}{s+2\pi\cdot 33.7}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 3.63\text{k}}} \quad \text{(corners: 33.7 Hz, 3.63 kHz)}$$
$$V_o(t) = 2 + 0.48\sin(2\pi\cdot 1000\,t - 0.235)\ \text{V}$$

Solution 1B.8 **Medium**

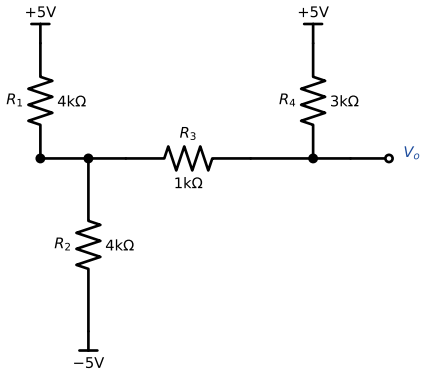


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies $-5\,\text{V}$, $+5\,\text{V}$. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1, C_2). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal $0\,\text{V}$ source: $(V_{th}, R_{th}) = (0\,\text{V},\, 0\,\Omega)$.
2. C_1 is a capacitor $\Rightarrow$ open at DC. No DC current can cross it, so everything on its left is disconnected: start fresh after it.
3. First DC path here: R_1 to the $+5\,\text{V}$ supply, so $(V_{th}, R_{th}) = (5\,\text{V},\ 4\,\text{k}\Omega)$.
4. Add R_2 to the $-5\,\text{V}$ supply. Divider between $V_a = 5\,\text{V}$ (through $R_a = 4\,\text{k}\Omega$) and $V_b = -5\,\text{V}$ (through $R_b = 4\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = -5 + (5 - (-5))\,\frac{4\,\text{k}}{4\,\text{k} + 4\,\text{k}} = 0\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 2\,\text{k}\Omega$$

5. Walk through R_3 ($1\,\text{k}\Omega$ at DC). A series resistor just adds to R_{th}: $(V_{th}, R_{th}) = (0\,\text{V},\ 3\,\text{k}\Omega)$.

6. Finally, collect everything at the output node V_o.
7. C_2 (at V_o) is a capacitor: open at DC, ignore it.
8. Add R_4 to the $+5\,\mathrm{V}$ supply. Divider between $V_a = 0\,\mathrm{V}$ (through $R_a = 3\,\mathrm{k\Omega}$) and $V_b = 5\,\mathrm{V}$ (through $R_b = 3\,\mathrm{k\Omega}$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 5 + (0 - (5))\,\frac{3\,\mathrm{k}}{3\,\mathrm{k} + 3\,\mathrm{k}} = 2.5\,\mathrm{V}, \qquad R_{th} = R_a \parallel R_b = 1.5\,\mathrm{k\Omega}$$

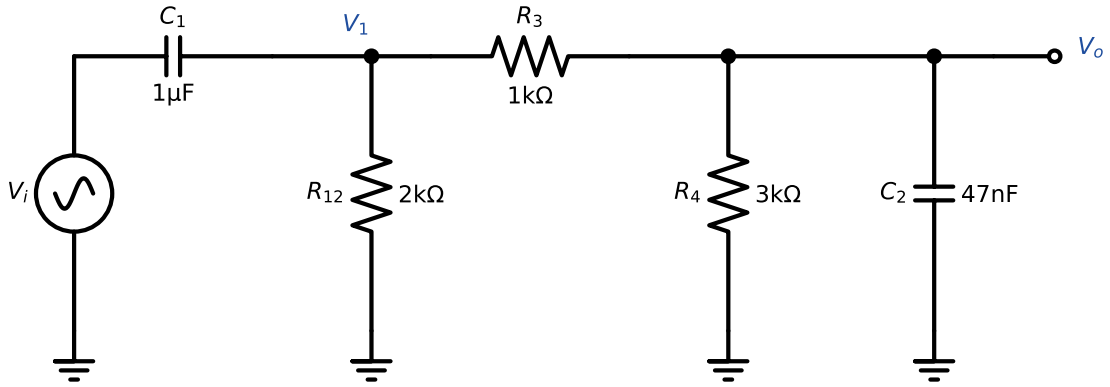
$$\boxed{V_{o,dc} = 2.5\,\mathrm{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1, R_2 both go from the same node to (AC) ground, so they are in parallel: $R_{12} = R_1 \parallel R_2 = 4\,\mathrm{k} \parallel 4\,\mathrm{k} = 2\,\mathrm{k\Omega}$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_4 \parallel \frac{1}{sC_2} = \frac{2.13 \times 10^7}{s + 7090}$$

$$Z_1 = R_{12} \parallel (R_3 + Z_o) = \frac{667\,s + 1.89 \times 10^7}{s + 14200}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^2 + 28400\,s}{s^2 + 29900\,s + 2.13 \times 10^7}$$

$$\frac{V_o}{V_1} = \frac{Z_o}{R_3 + Z_o} = \frac{21300}{s + 28400}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{21300\, s}{s^2 + 29900\, s + 2.13 \times 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 29900\, s + 2.13 \times 10^7$; quadratic formula:

$$s = \frac{-29900 \pm \sqrt{(29900)^2 - 4(2.13 \times 10^7)}}{2} = \frac{-29900 \pm 28400}{2} \Rightarrow s = -730,\ -29100$$

Factored form (leading coefficient in front):

$$H(s) = 21300\, \frac{s}{(s + 730)(s + 29100)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.73\, \frac{s}{s + 2\pi \cdot 116}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 4.64\text{k}}}, \qquad K = 0.73$$

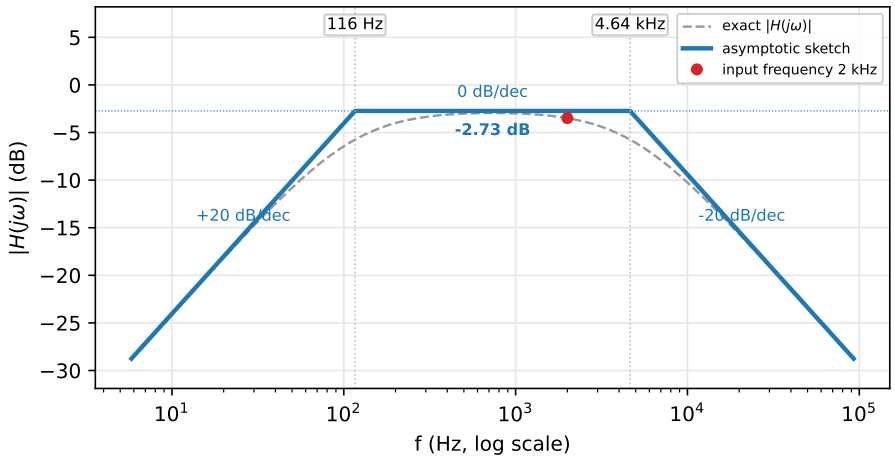
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	730	116 Hz	−20 dB/dec
pole	29100	4.64 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.001\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 116 Hz	+20 dB/dec	-2.73 dB at end
116 Hz – 4.64 kHz	0 dB/dec (flat)	-2.73 dB at start, -2.73 dB at end
$f >$ 4.64 kHz	−20 dB/dec	-2.73 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 0.73 = -2.73$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi \cdot 2000\,t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.73	0.7302	0
$\dfrac{s}{s+2\pi\cdot 116}$	0.9983	0.058
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 4.64\text{k}}}$	0.9182	−0.407
total	**0.6694** (product)	**−0.349** (sum)

$$|H(j\omega_{in})| = 0.669, \qquad \angle H(j\omega_{in}) = -0.349 \text{ rad}$$
$$v_{o,ac}(t) = 0.669 \times 0.5\,\sin(2\pi \cdot 2000\,t - 0.349) = 0.335\,\sin(2\pi \cdot 2000\,t - 0.349)$$

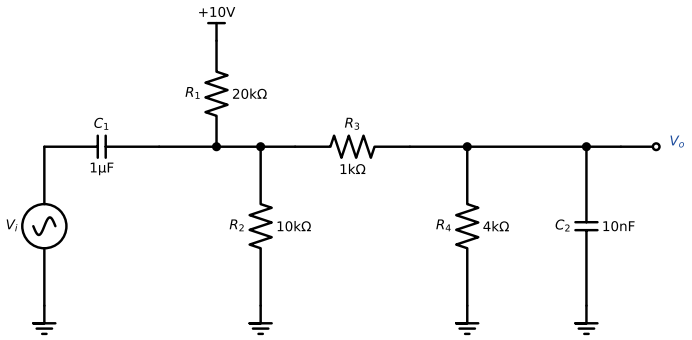
Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = 2.5 + 0.335\,\sin(2\pi \cdot 2000\,t - 0.349)\ \text{V}}$$

Answer

$$H(s) = 0.73\,\frac{s}{s+2\pi\cdot 116}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 4.64\text{k}}} \quad \text{(corners: 116 Hz, 4.64 kHz)}$$
$$V_o(t) = 2.5 + 0.335\,\sin(2\pi \cdot 2000\,t - 0.349)\ \text{V}$$

Solution 1B.9 **Medium**

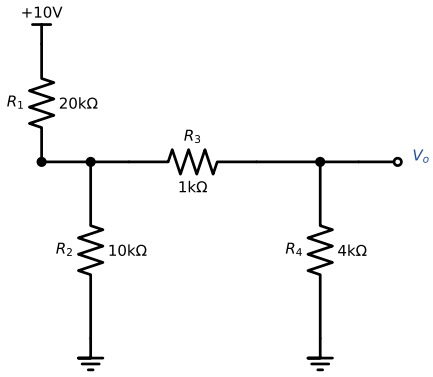


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies +10 V. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1, C_2). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal 0 V source: $(V_{th}, R_{th}) = (0\,\text{V},\, 0\,\Omega)$.
2. C_1 is a capacitor $\Rightarrow$ open at DC. No DC current can cross it, so everything on its left is disconnected: start fresh after it.
3. First DC path here: R_1 to the +10 V supply, so $(V_{th}, R_{th}) = (10\,\text{V},\ 20\,\text{k}\Omega)$.
4. Add R_2 to ground. Divider between $V_a = 10$ V (through $R_a = 20\,\text{k}\Omega$) and $V_b = 0$ V (through $R_b = 10\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 0 + (10 - (0))\,\frac{10\,\text{k}}{20\,\text{k} + 10\,\text{k}} = 3.33\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 6.67\,\text{k}\Omega$$

5. Walk through R_3 (1 kΩ at DC). A series resistor just adds to R_{th}: $(V_{th}, R_{th}) = (3.33\,\text{V},\ 7.67\,\text{k}\Omega)$.

6. Finally, collect everything at the output node V_o.

7. Add R_4 to ground. Divider between $V_a = 3.33\,\text{V}$ (through $R_a = 7.67\,\text{k}\Omega$) and $V_b = 0\,\text{V}$ (through $R_b = 4\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 0 + (3.33 - (0))\,\frac{4\,\text{k}}{7.67\,\text{k} + 4\,\text{k}} = 1.14\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 2.63\,\text{k}\Omega$$

8. C_2 (at V_o) is a capacitor: open at DC, ignore it.

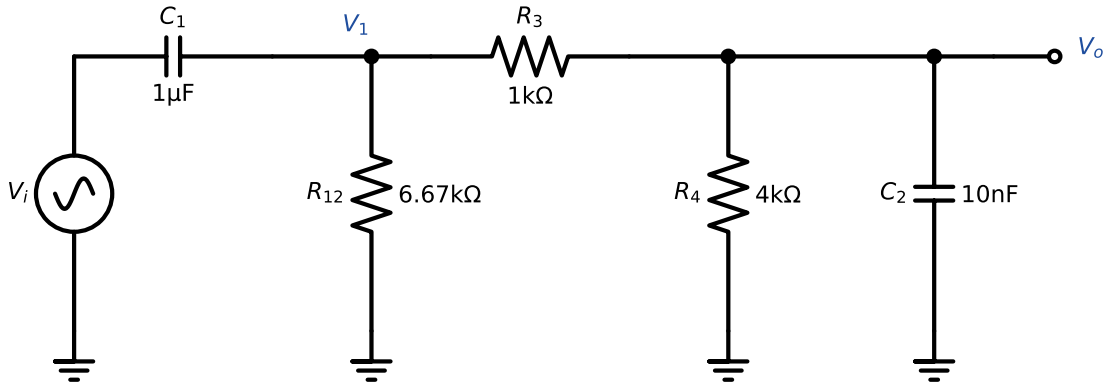
$$\boxed{V_{o,dc} = 1.14\,\text{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1, R_2 both go from the same node to (AC) ground, so they are in parallel: $R_{12} = R_1 \parallel R_2 = 20\,\text{k} \parallel 10\,\text{k} = 6.67\,\text{k}\Omega$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_4 \parallel \frac{1}{sC_2} = \frac{10^8}{s + 25000}$$

$$Z_1 = R_{12} \parallel (R_3 + Z_o) = \frac{870\,s + 1.09 \times 10^8}{s + 38000}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^2 + 1.25 \times 10^5\, s}{s^2 + 1.26 \times 10^5\, s + 4.38 \times 10^7}$$
$$\frac{V_o}{V_1} = \frac{Z_o}{R_3 + Z_o} = \frac{10^5}{s + 1.25 \times 10^5}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{10^5\, s}{s^2 + 1.26 \times 10^5\, s + 4.38 \times 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 1.26 \times 10^5\, s + 4.38 \times 10^7$; quadratic formula:

$$s = \frac{-1.26 \times 10^5 \pm \sqrt{(1.26 \times 10^5)^2 - 4(4.38 \times 10^7)}}{2} = \frac{-1.26 \times 10^5 \pm 1.25 \times 10^5}{2} \Rightarrow s = -348,\ -1.26{\times}10^5$$

Factored form (leading coefficient in front):

$$H(s) = 10^5\, \frac{s}{(s + 348)(s + 1.26 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.795\, \frac{s}{s + 2\pi \cdot 55.3}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 20\text{k}}}, \qquad K = 0.795$$

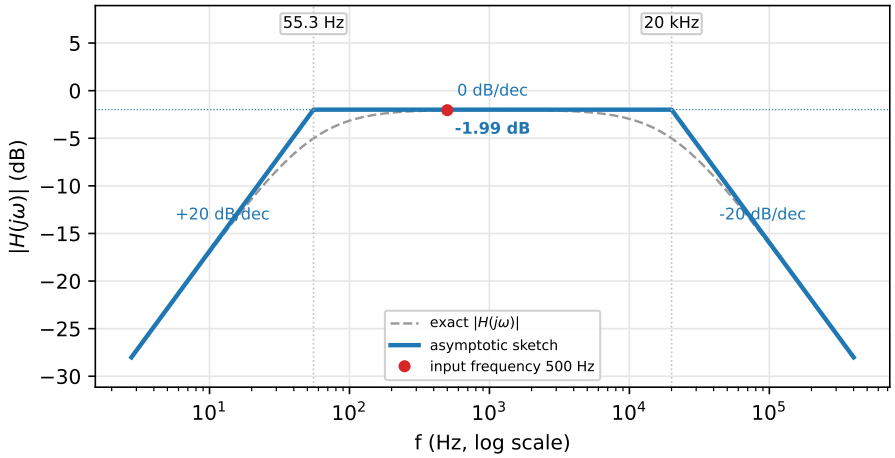
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	348	55.3 Hz	−20 dB/dec
pole	1.26×10^5	20 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.00229\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 55.3$ Hz	+20 dB/dec	-1.99 dB at end
55.3 Hz – 20 kHz	0 dB/dec (flat)	-1.99 dB at start, -1.99 dB at end
$f > 20$ kHz	−20 dB/dec	-1.99 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}0.795 = -1.99\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi\cdot 500\,t)$, so $\omega_{in} = 2\pi\cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.795	0.7949	0
$\dfrac{s}{s+2\pi\cdot 55.3}$	0.9939	0.11
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$	0.9997	-0.025
total	**0.7898** (product)	**0.0853** (sum)

$$|H(j\omega_{in})| = 0.79, \qquad \angle H(j\omega_{in}) = 0.0853\ \text{rad}$$
$$v_{o,ac}(t) = 0.79\times 1\sin(2\pi\cdot 500\,t + 0.0853) = 0.79\sin(2\pi\cdot 500\,t + 0.0853)$$

Add the DC part from Step 2 (superposition):

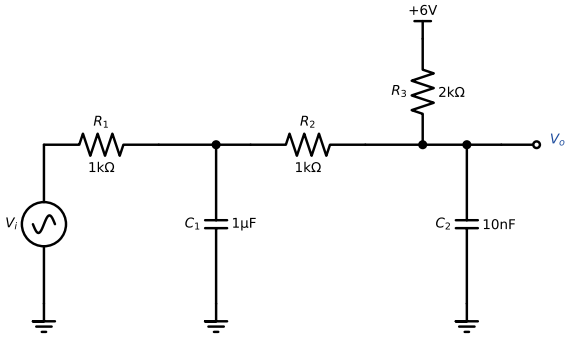
$$\boxed{V_o(t) = 1.14 + 0.79\sin(2\pi\cdot 500\,t + 0.0853)\ \text{V}}$$

Answer

$$H(s) = 0.795\,\frac{s}{s+2\pi\cdot 55.3}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}} \quad \text{(corners: 55.3 Hz, 20 kHz)}$$

$$V_o(t) = 1.14 + 0.79\,\sin\left(2\pi \cdot 500\,t + 0.0853\right)\ \text{V}$$

Solution 1B.10 **Medium**

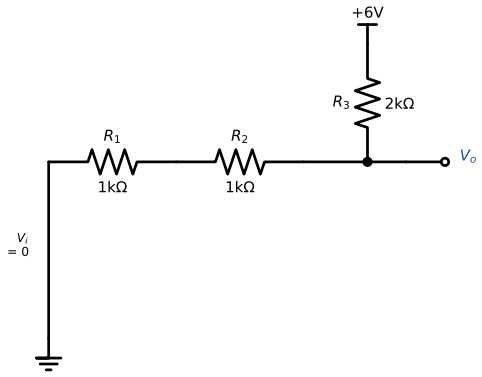


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies +6 V. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1, C_2). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal 0 V source: $(V_{th}, R_{th}) = (0\,\text{V},\, 0\,\Omega)$.
2. Walk through R_1 (1 kΩ at DC). A series resistor just adds to R_{th}: $(V_{th}, R_{th}) = (0\,\text{V},\ 1\,\text{k}\Omega)$.
3. C_1 (at V_1) is a capacitor: open at DC, ignore it.
4. Walk through R_2 (1 kΩ at DC). A series resistor just adds to R_{th}: $(V_{th}, R_{th}) = (0\,\text{V},\ 2\,\text{k}\Omega)$.
5. Finally, collect everything at the output node V_o.
6. Add R_3 to the +6 V supply. Divider between $V_a = 0\,\text{V}$ (through $R_a = 2\,\text{k}\Omega$) and $V_b = 6\,\text{V}$ (through $R_b = 2\,\text{k}\Omega$):

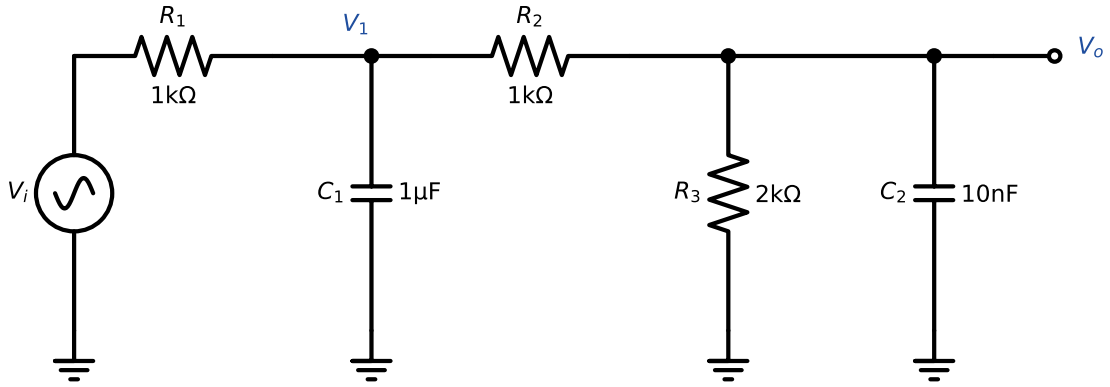
$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 6 + (0 - (6))\,\frac{2\,\text{k}}{2\,\text{k} + 2\,\text{k}} = 3\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 1\,\text{k}\Omega$$

7. C_2 (at V_o) is a capacitor: open at DC, ignore it.

$$\boxed{V_{o,dc} = 3\,\text{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_3 \parallel \frac{1}{sC_2} = \frac{10^8}{s + 50000}$$
$$Z_1 = \frac{1}{sC_1} \parallel (R_2 + Z_o) = \frac{10^6\, s + 1.5 \times 10^{11}}{s^2 + 1.51 \times 10^5\, s + 5 \times 10^7}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{R_1 + Z_1} = \frac{1000\, s + 1.5 \times 10^8}{s^2 + 1.52 \times 10^5\, s + 2 \times 10^8}$$
$$\frac{V_o}{V_1} = \frac{Z_o}{R_2 + Z_o} = \frac{10^5}{s + 1.5 \times 10^5}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{10^8}{s^2 + 1.52 \times 10^5\, s + 2 \times 10^8}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 1.52 \times 10^5\, s + 2 \times 10^8$; quadratic formula:

$$s = \frac{-1.52 \times 10^5 \pm \sqrt{(1.52 \times 10^5)^2 - 4(2 \times 10^8)}}{2} = \frac{-1.52 \times 10^5 \pm 1.49 \times 10^5}{2} \Rightarrow s = -1330,\ -1.51{\times}10^5$$

Factored form (leading coefficient in front):

$$H(s) = 10^8 \, \frac{1}{(s + 1330)(s + 1.51 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.5 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 211}} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 24\text{k}}}, \qquad K = 0.5$$

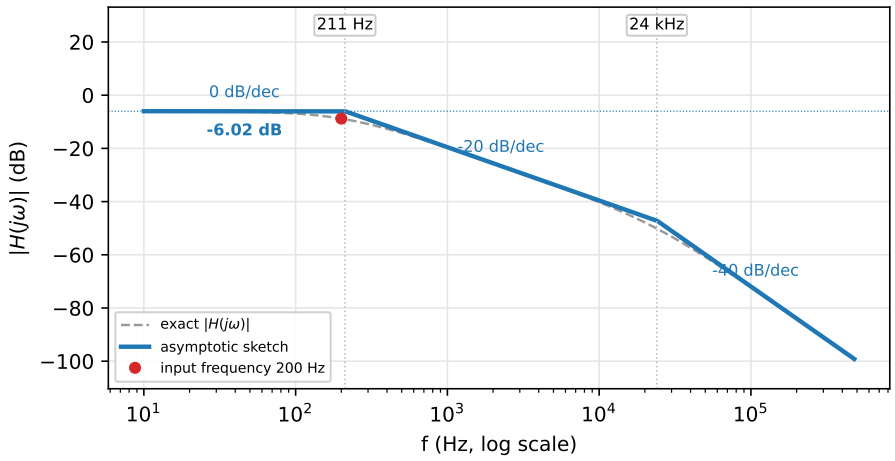
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1330	211 Hz	−20 dB/dec
pole	1.51×10^5	24 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 0.5 = -6.02\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 211$ Hz	0 dB/dec (flat)	-6.02 dB at end
211 Hz – 24 kHz	−20 dB/dec	-6.02 dB at start, -47.1 dB at end
$f > 24$ kHz	−40 dB/dec	-47.1 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 0.5 = -6.02\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi \cdot 200\,t)$, so $\omega_{in} = 2\pi \cdot 200 = 1260\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.5	0.5	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 211}}$	0.7262	−0.758
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 24\text{k}}}$	1	−0.00834
total	**0.3631** (product)	**−0.766** (sum)

$$|H(j\omega_{in})| = 0.363, \qquad \angle H(j\omega_{in}) = -0.766 \text{ rad}$$
$$v_{o,ac}(t) = 0.363 \times 1 \sin(2\pi \cdot 200\,t - 0.766) = 0.363 \sin(2\pi \cdot 200\,t - 0.766)$$

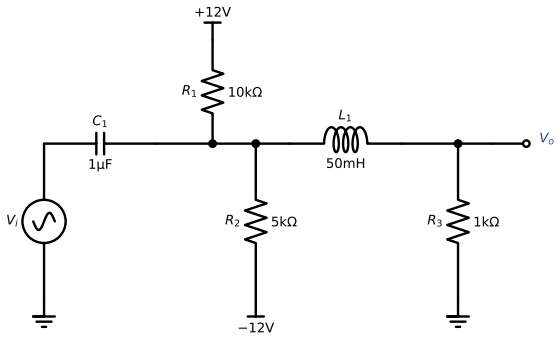
Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = 3 + 0.363 \sin(2\pi \cdot 200\,t - 0.766)\ \text{V}}$$

Answer

$$H(s) = 0.5\,\frac{1}{1+\dfrac{s}{2\pi \cdot 211}}\,\frac{1}{1+\dfrac{s}{2\pi \cdot 24\text{k}}} \quad \text{(corners: 211 Hz, 24 kHz)}$$
$$V_o(t) = 3 + 0.363 \sin(2\pi \cdot 200\,t - 0.766)\ \text{V}$$

Solution 1B.11 **Medium**

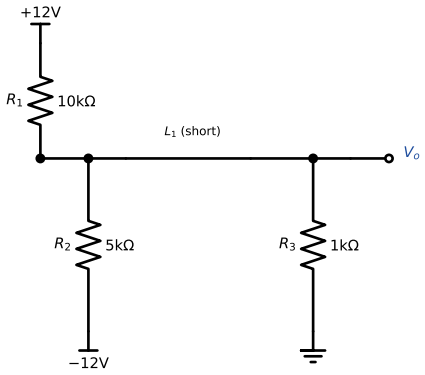


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies $-12\,\text{V}$, $+12\,\text{V}$. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1); every inductor is a short circuit (L_1). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal $0\,\text{V}$ source: $(V_{th}, R_{th}) = (0\,\text{V},\, 0\,\Omega)$.
2. C_1 is a capacitor $\Rightarrow$ open at DC. No DC current can cross it, so everything on its left is disconnected: start fresh after it.
3. First DC path here: R_1 to the $+12\,\text{V}$ supply, so $(V_{th}, R_{th}) = (12\,\text{V},\ 10\,\text{k}\Omega)$.
4. Add R_2 to the $-12\,\text{V}$ supply. Divider between $V_a = 12\,\text{V}$ (through $R_a = 10\,\text{k}\Omega$) and $V_b = -12\,\text{V}$ (through $R_b = 5\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = -12 + (12 - (-12))\,\frac{5\,\text{k}}{10\,\text{k} + 5\,\text{k}} = -4\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 3.33\,\text{k}\Omega$$

5. L_1 is a short at DC (inductor), so the pair is unchanged.

6. Finally, collect everything at the output node V_o.

7. Add R_3 to ground. Divider between $V_a = -4\,\mathrm{V}$ (through $R_a = 3.33\,\mathrm{k\Omega}$) and $V_b = 0\,\mathrm{V}$ (through $R_b = 1\,\mathrm{k\Omega}$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 0 + (-4 - (0))\,\frac{1\,\mathrm{k}}{3.33\,\mathrm{k} + 1\,\mathrm{k}} = -923\,\mathrm{mV}, \qquad R_{th} = R_a \parallel R_b = 769\,\Omega$$

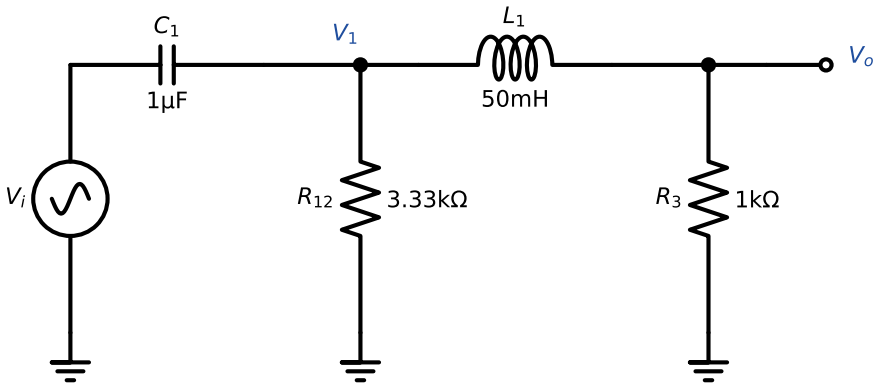
$$\boxed{V_{o,dc} = -923\,\mathrm{mV}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1, R_2 both go from the same node to (AC) ground, so they are in parallel: $R_{12} = R_1 \parallel R_2 = 10\,\mathrm{k} \parallel 5\,\mathrm{k} = 3.33\,\mathrm{k\Omega}$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_{12} \parallel (sL_1 + R_3) = \frac{3330\,s + 6.67 \times 10^7}{s + 86700}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^2 + 20000\,s}{s^2 + 20300\,s + 2.6 \times 10^7}$$

$$\frac{V_o}{V_1} = \frac{R_3}{sL_1 + R_3} = \frac{20000}{s + 20000}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{20000\, s}{s^2 + 20300\, s + 2.6 \times 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 20300\, s + 2.6 \times 10^7$; quadratic formula:

$$s = \frac{-20300 \pm \sqrt{(20300)^2 - 4(2.6 \times 10^7)}}{2} = \frac{-20300 \pm 17600}{2} \Rightarrow s = -1370,\ -18900$$

Factored form (leading coefficient in front):

$$H(s) = 20000\, \frac{s}{(s + 1370)(s + 18900)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1.06\, \frac{s}{s + 2\pi \cdot 219}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 3.01\text{k}}}, \qquad K = 1.06$$

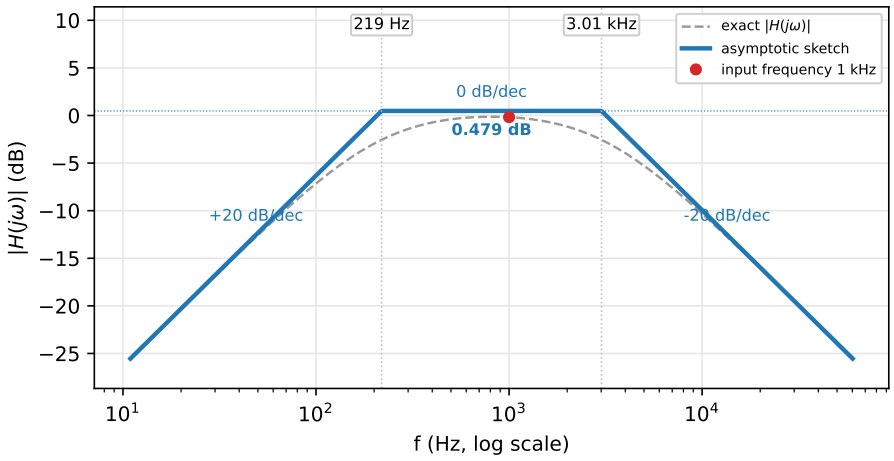
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1370	219 Hz	−20 dB/dec
pole	18900	3.01 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 7.69 \times 10^{-4}\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 219 Hz	+20 dB/dec	0.479 dB at end
219 Hz – 3.01 kHz	0 dB/dec (flat)	0.479 dB at start, 0.479 dB at end
$f >$ 3.01 kHz	−20 dB/dec	0.479 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 1.06 = 0.479$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 2\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1.06	1.057	0
$\dfrac{s}{s+2\pi\cdot 219}$	0.9769	0.215
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 3.01\text{k}}}$	0.9491	−0.321
total	**0.9798** (product)	**−0.105** (sum)

$$|H(j\omega_{in})| = 0.98, \qquad \angle H(j\omega_{in}) = -0.105 \text{ rad}$$
$$v_{o,ac}(t) = 0.98 \times 2\sin(2\pi\cdot 1000\,t - 0.105) = 1.96\sin(2\pi\cdot 1000\,t - 0.105)$$

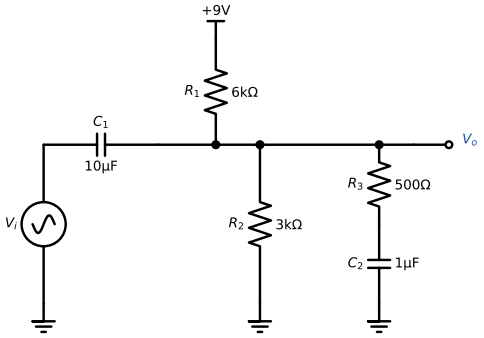
Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = -0.923 + 1.96\sin(2\pi\cdot 1000\,t - 0.105)\ \text{V}}$$

Answer

$$H(s) = 1.06\,\frac{s}{s+2\pi\cdot 219}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 3.01\text{k}}} \quad \text{(corners: 219 Hz, 3.01 kHz)}$$
$$V_o(t) = -0.923 + 1.96\sin(2\pi\cdot 1000\,t - 0.105)\ \text{V}$$

Solution 1B.12 **Medium**

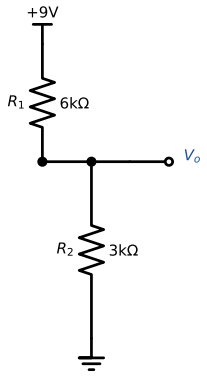


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies +9 V. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1, C_2). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal 0 V source: $(V_{th}, R_{th}) = (0\,\text{V},\, 0\,\Omega)$.
2. C_1 is a capacitor $\Rightarrow$ open at DC. No DC current can cross it, so everything on its left is disconnected: start fresh after it.
3. Finally, collect everything at the output node V_o.
4. First DC path here: R_1 to the +9 V supply, so $(V_{th}, R_{th}) = (9\,\text{V},\ 6\,\text{k}\Omega)$.
5. Add R_2 to ground. Divider between $V_a = 9$ V (through $R_a = 6\,\text{k}\Omega$) and $V_b = 0$ V (through $R_b = 3\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 0 + (9 - (0))\,\frac{3\,\text{k}}{6\,\text{k} + 3\,\text{k}} = 3\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 2\,\text{k}\Omega$$

6. $R_3 + C_2$ (at V_o) contains a capacitor: open at DC, ignore it.

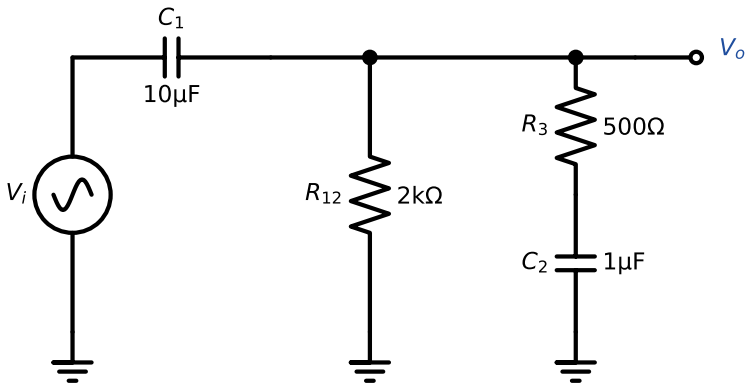
$$\boxed{V_{o,dc} = 3\,\mathrm{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1, R_2 both go from the same node to (AC) ground, so they are in parallel: $R_{12} = R_1 \parallel R_2 = 6\,\mathrm{k} \parallel 3\,\mathrm{k} = 2\,\mathrm{k\Omega}$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_{12} \parallel \left(R_3 + \frac{1}{sC_2}\right) = \frac{400\,s + 8 \times 10^5}{s + 400}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{\frac{1}{sC_1} + Z_o} = \frac{s^2 + 2000\,s}{s^2 + 2250\,s + 10^5}$$

So:

$$H(s) = \frac{s^2 + 2000\,s}{s^2 + 2250\,s + 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Numerator (after taking out the s factor): $s + 2000$; $s + 2000 = 0 \Rightarrow s = -2000$.

- Denominator: $s^2 + 2250\,s + 10^5$; quadratic formula:

$$s = \frac{-2250 \pm \sqrt{(2250)^2 - 4(10^5)}}{2} = \frac{-2250 \pm 2160}{2} \Rightarrow s = -45.4,\ -2200$$

Factored form (leading coefficient in front):

$$H(s) = 1\,\frac{s(s+2000)}{(s+45.4)(s+2200)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

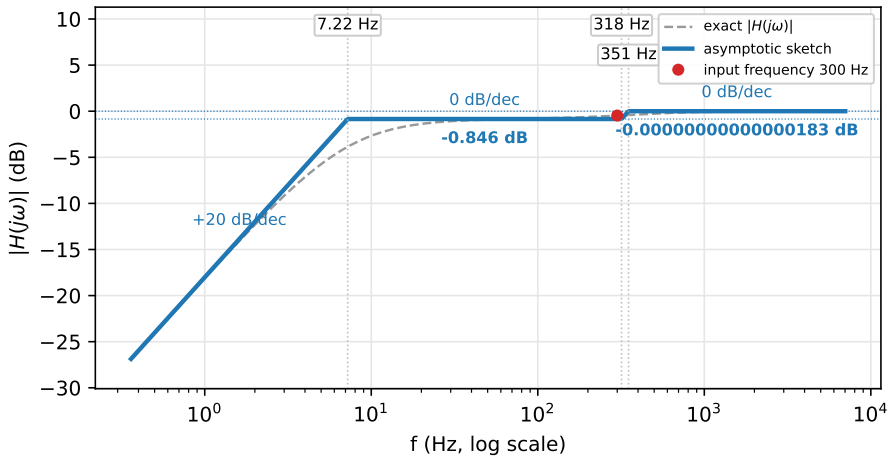
$$H(s) = 0.907\,\frac{s}{s + 2\pi\cdot 7.22}\left(1+\frac{s}{2\pi\cdot 318}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 351}}, \qquad K = 0.907$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	45.4	7.22 Hz	−20 dB/dec
zero	2000	318 Hz	+20 dB/dec
pole	2200	351 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.02\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 7.22$ Hz	+20 dB/dec	-0.846 dB at end
7.22 Hz – 318 Hz	0 dB/dec (flat)	-0.846 dB at start, -0.846 dB at end
318 Hz – 351 Hz	+20 dB/dec	-0.846 dB at start, 0 dB at end
$f > 351$ Hz	0 dB/dec (flat)	0 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.4\sin(2\pi \cdot 300\,t)$, so $\omega_{in} = 2\pi \cdot 300 = 1880$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.907	0.9072	0
$\dfrac{s}{s+2\pi\cdot 7.22}$	0.9997	0.0241
$\left(1+\dfrac{s}{2\pi\cdot 318}\right)$	1.374	0.756
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 351}}$	0.7601	−0.707
total	**0.9472** (product)	**0.0725** (sum)

$$|H(j\omega_{in})| = 0.947, \qquad \angle H(j\omega_{in}) = 0.0725 \text{ rad}$$
$$v_{o,ac}(t) = 0.947 \times 0.4\,\sin(2\pi \cdot 300\,t + 0.0725) = 0.379\,\sin(2\pi \cdot 300\,t + 0.0725)$$

Add the DC part from Step 2 (superposition):

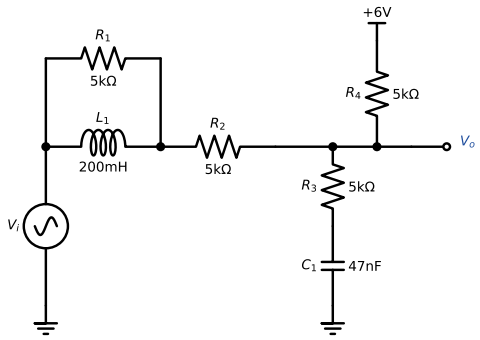
$$\boxed{V_o(t) = 3 + 0.379\,\sin(2\pi \cdot 300\,t + 0.0725)\ \text{V}}$$

Answer

$$H(s) = 0.907\,\frac{s}{s+2\pi\cdot 7.22}\left(1+\frac{s}{2\pi\cdot 318}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 351}} \quad \text{(corners: 7.22 Hz, 318 Hz, 351 Hz)}$$

$$V_o(t) = 3 + 0.379\,\sin(2\pi \cdot 300\,t + 0.0725)\ \text{V}$$

Solution 1B.13 **Hard**

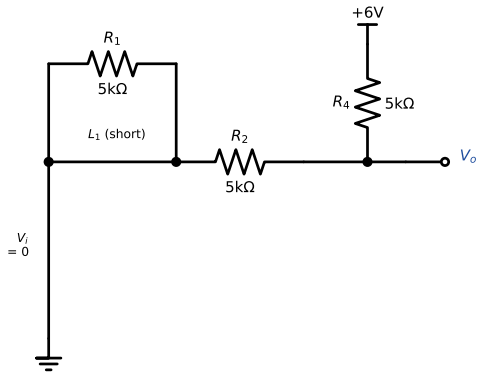


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies +6 V. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1); every inductor is a short circuit (L_1). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

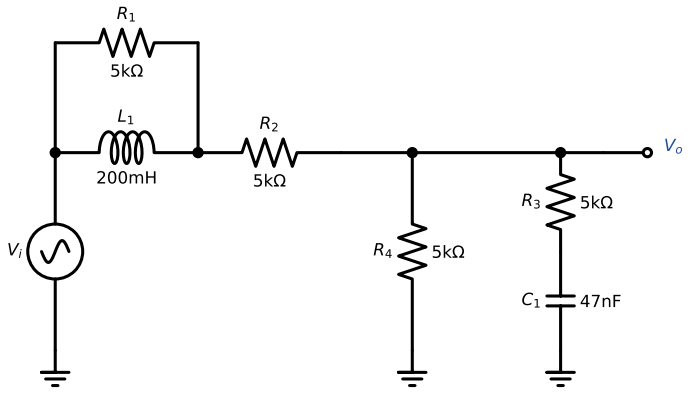
1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal 0 V source: $(V_{th}, R_{th}) = (0\,\text{V},\ 0\,\Omega)$.
2. $L_1 \parallel R_1$ is a short at DC (inductor), so the pair is unchanged.
3. Walk through R_2 (5 kΩ at DC). A series resistor just adds to R_{th}: $(V_{th}, R_{th}) = (0\,\text{V},\ 5\,\text{k}\Omega)$.
4. Finally, collect everything at the output node V_o.
5. $R_3 + C_1$ (at V_o) contains a capacitor: open at DC, ignore it.
6. Add R_4 to the +6 V supply. Divider between $V_a = 0\,\text{V}$ (through $R_a = 5\,\text{k}\Omega$) and $V_b = 6\,\text{V}$ (through $R_b = 5\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 6 + (0 - (6))\,\frac{5\,\text{k}}{5\,\text{k} + 5\,\text{k}} = 3\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 2.5\,\text{k}\Omega$$

$$\boxed{V_{o,dc} = 3\,\mathrm{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_4 \parallel \left(R_3 + \frac{1}{sC_1}\right) = \frac{2500\,s + 1.06 \times 10^7}{s + 2130}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{(sL_1 \parallel R_1) + R_2 + Z_o} = \frac{0.2\,s^2 + 5850\,s + 2.13 \times 10^7}{s^2 + 17600\,s + 4.26 \times 10^7}$$

So:

$$H(s) = \frac{0.2\,s^2 + 5850\,s + 2.13 \times 10^7}{s^2 + 17600\,s + 4.26 \times 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $0.2\,s^2 + 5850\,s + 2.13 \times 10^7$; divide by the leading coefficient 0.2 (this does not move the roots): $s^2 + 29300\,s + 1.06 \times 10^8$; quadratic formula:

$$s = \frac{-29300 \pm \sqrt{(29300)^2 - 4(1.06 \times 10^8)}}{2} = \frac{-29300 \pm 20700}{2} \Rightarrow s = -4260,\ -25000$$

- Denominator: $s^2 + 17600\,s + 4.26 \times 10^7$; quadratic formula:

$$s = \frac{-17600 \pm \sqrt{(17600)^2 - 4(4.26 \times 10^7)}}{2} = \frac{-17600 \pm 11700}{2} \Rightarrow s = -2910,\ -14600$$

Factored form (leading coefficient in front):

$$H(s) = 0.2\,\frac{(s+4260)(s+25000)}{(s+2910)(s+14600)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

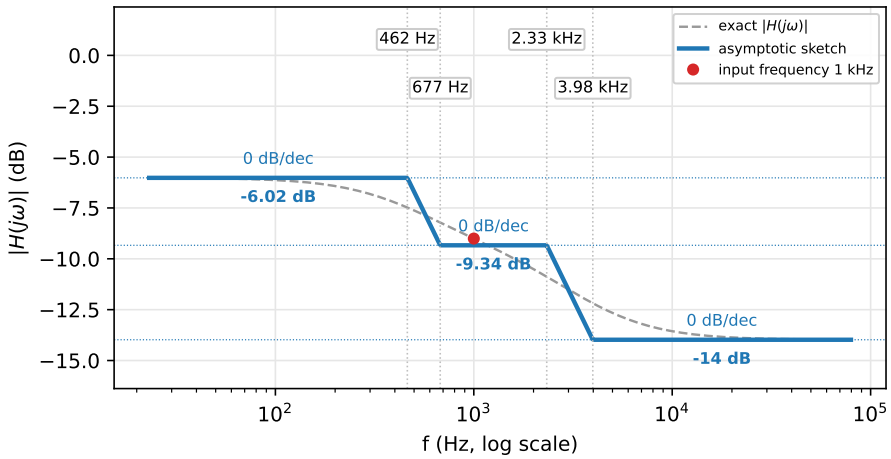
$$H(s) = 0.5\,\frac{1}{1+\dfrac{s}{2\pi\cdot 462}}\left(1+\frac{s}{2\pi\cdot 677}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 2.33\text{k}}}\left(1+\frac{s}{2\pi\cdot 3.98\text{k}}\right), \qquad K = 0.5$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	2910	462 Hz	−20 dB/dec
zero	4260	677 Hz	+20 dB/dec
pole	14600	2.33 kHz	−20 dB/dec
zero	25000	3.98 kHz	+20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 0.5 = -6.02\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 462$ Hz	0 dB/dec (flat)	-6.02 dB at end
462 Hz – 677 Hz	−20 dB/dec	-6.02 dB at start, -9.34 dB at end
677 Hz – 2.33 kHz	0 dB/dec (flat)	-9.34 dB at start, -9.34 dB at end
2.33 kHz – 3.98 kHz	−20 dB/dec	-9.34 dB at start, -14 dB at end
$f > 3.98$ kHz	0 dB/dec (flat)	-14 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1.5 \sin\left(2\pi \cdot 1000\, t + \frac{\pi}{4}\right)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.5	0.5	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 462}}$	0.4197	−1.14
$\left(1+\dfrac{s}{2\pi \cdot 677}\right)$	1.783	0.975
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 2.33\text{k}}}$	0.919	−0.405
$\left(1+\dfrac{s}{2\pi \cdot 3.98\text{k}}\right)$	1.031	0.246
total	**0.3546** (product)	**−0.321** (sum)

$$|H(j\omega_{in})| = 0.355, \qquad \angle H(j\omega_{in}) = -0.321 \text{ rad}$$
$$v_{o,ac}(t) = 0.355 \times 1.5 \sin\left(2\pi \cdot 1000\, t + 0.464\right) = 0.532 \sin\left(2\pi \cdot 1000\, t + 0.464\right)$$

(phase: $\theta + \angle H = \frac{\pi}{4} + (-0.321) = 0.464$ rad.) Add the DC part from Step 2 (superposition):

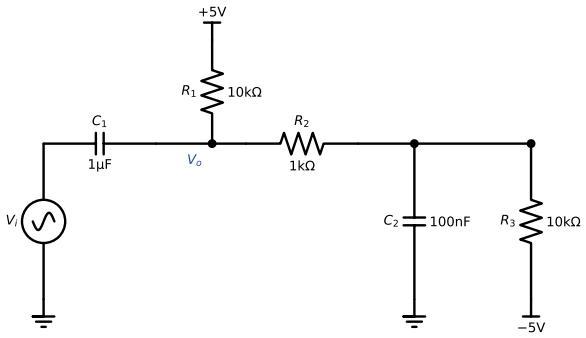
$$\boxed{V_o(t) = 3 + 0.532 \sin\left(2\pi \cdot 1000\, t + 0.464\right) \text{ V}}$$

Answer

$H(s) = 0.5 \dfrac{1}{1+\dfrac{s}{2\pi \cdot 462}} \left(1+\dfrac{s}{2\pi \cdot 677}\right) \dfrac{1}{1+\dfrac{s}{2\pi \cdot 2.33\text{k}}} \left(1+\dfrac{s}{2\pi \cdot 3.98\text{k}}\right)$ (corners: 462 Hz, 677 Hz, 2.33 kHz, 3.98 kHz)

$V_o(t) = 3 + 0.532 \sin\left(2\pi \cdot 1000\, t + 0.464\right)$ V

Solution 1B.14 **Hard**

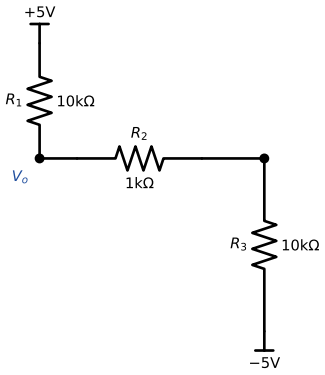


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies $-5\,\text{V}$, $+5\,\text{V}$. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1, C_2). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal $0\,\text{V}$ source: $(V_{th}, R_{th}) = (0\,\text{V},\, 0\,\Omega)$.
2. C_1 is a capacitor $\Rightarrow$ open at DC. No DC current can cross it, so everything on its left is disconnected: start fresh after it.
3. Now reduce the part to the *right* of V_o the same way, starting at the far right end and moving toward V_o.
4. C_2 (at V_1) is a capacitor: open at DC, ignore it.
5. First DC path here: R_3 to the $-5\,\text{V}$ supply, so $(V_{th}, R_{th}) = (-5\,\text{V},\ 10\,\text{k}\Omega)$.
6. Walk through R_2 ($1\,\text{k}\Omega$ at DC). A series resistor just adds to R_{th}: $(V_{th}, R_{th}) = (-5\,\text{V},\ 11\,\text{k}\Omega)$.
7. Finally, collect everything at the output node V_o.

8. First DC path here: R_1 to the $+5\,\text{V}$ supply, so $(V_{th}, R_{th}) = (5\,\text{V},\ 10\,\text{k}\Omega)$.

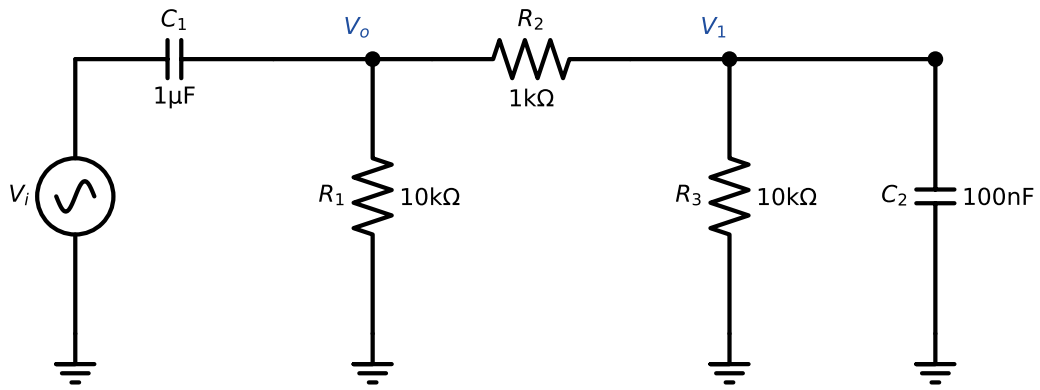
9. V_o sits between the left pair $(5\,\text{V},\ 10\,\text{k}\Omega)$ and the right pair $(-5\,\text{V},\ 11\,\text{k}\Omega)$:

$$V_{o,dc} = -5 + (5 - (-5))\,\frac{11\,\text{k}}{10\,\text{k} + 11\,\text{k}} = 238\,\text{mV}$$

$$\boxed{V_{o,dc} = 238\,\text{mV}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_3 \parallel \frac{1}{sC_2} = \frac{10^7}{s + 1000}$$

$$Z_o = R_1 \parallel (R_2 + Z_1) = \frac{909\,s + 10^7}{s + 1910}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{\frac{1}{sC_1} + Z_o} = \frac{s^2 + 11000\,s}{s^2 + 12100\,s + 2.1 \times 10^6}$$

So:

$$H(s) = \frac{s^2 + 11000\,s}{s^2 + 12100\,s + 2.1 \times 10^6}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Numerator (after taking out the s factor): $s + 11000$; $s + 11000 = 0 \Rightarrow s = -11000$.
- Denominator: $s^2 + 12100\,s + 2.1 \times 10^6$; quadratic formula:

$$s = \frac{-12100 \pm \sqrt{(12100)^2 - 4(2.1 \times 10^6)}}{2} = \frac{-12100 \pm 11700}{2} \Rightarrow s = -176,\ -11900$$

Factored form (leading coefficient in front):

$$H(s) = 1\,\frac{s(s + 11000)}{(s + 176)(s + 11900)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

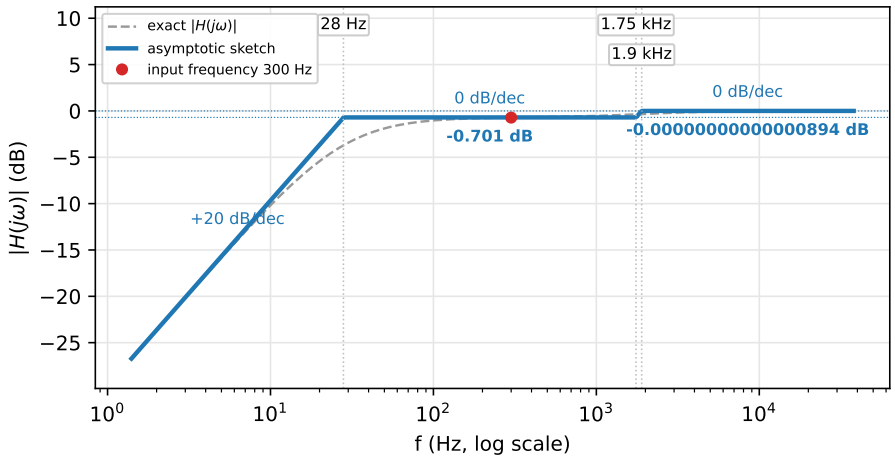
$$H(s) = 0.923\,\frac{s}{s + 2\pi \cdot 28}\left(1 + \frac{s}{2\pi \cdot 1.75\text{k}}\right)\frac{1}{1 + \dfrac{s}{2\pi \cdot 1.9\text{k}}}, \qquad K = 0.923$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	176	28 Hz	−20 dB/dec
zero	11000	1.75 kHz	+20 dB/dec
pole	11900	1.9 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.00524\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 28 Hz	+20 dB/dec	-0.701 dB at end
28 Hz – 1.75 kHz	0 dB/dec (flat)	-0.701 dB at start, -0.701 dB at end
1.75 kHz – 1.9 kHz	+20 dB/dec	-0.701 dB at start, 0 dB at end
$f >$ 1.9 kHz	0 dB/dec (flat)	0 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi \cdot 300\,t)$, so $\omega_{in} = 2\pi \cdot 300 = 1880\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.923	0.9225	0
$\dfrac{s}{s+2\pi\cdot 28}$	0.9957	0.0932
$\left(1+\dfrac{s}{2\pi\cdot 1.75\text{k}}\right)$	1.015	0.17
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.9\text{k}}}$	0.9877	-0.157
total	**0.9205** (product)	**0.106** (sum)

$$|H(j\omega_{in})| = 0.92, \qquad \angle H(j\omega_{in}) = 0.106 \text{ rad}$$
$$v_{o,ac}(t) = 0.92 \times 0.5\sin(2\pi\cdot 300\,t + 0.106) = 0.46\sin(2\pi\cdot 300\,t + 0.106)$$

Add the DC part from Step 2 (superposition):

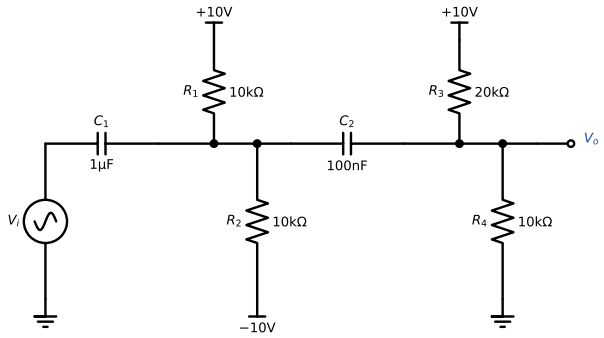
$$\boxed{V_o(t) = 0.238 + 0.46\sin(2\pi\cdot 300\,t + 0.106)\ \text{V}}$$

Answer

$$H(s) = 0.923\,\frac{s}{s+2\pi\cdot 28}\left(1+\frac{s}{2\pi\cdot 1.75\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 1.9\text{k}}} \quad \text{(corners: 28 Hz, 1.75 kHz, 1.9 kHz)}$$

$$V_o(t) = 0.238 + 0.46\,\sin\left(2\pi \cdot 300\,t + 0.106\right)\ \text{V}$$

Solution 1B.15 **Hard**

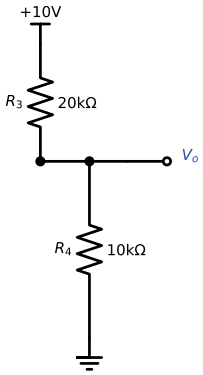


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies $-10\,\text{V}$, $+10\,\text{V}$. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1, C_2). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal $0\,\text{V}$ source: $(V_{th}, R_{th}) = (0\,\text{V},\, 0\,\Omega)$.
2. C_1 is a capacitor $\Rightarrow$ open at DC. No DC current can cross it, so everything on its left is disconnected: start fresh after it.
3. First DC path here: R_1 to the $+10\,\text{V}$ supply, so $(V_{th}, R_{th}) = (10\,\text{V},\ 10\,\text{k}\Omega)$.
4. Add R_2 to the $-10\,\text{V}$ supply. Divider between $V_a = 10\,\text{V}$ (through $R_a = 10\,\text{k}\Omega$) and $V_b = -10\,\text{V}$ (through $R_b = 10\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = -10 + (10 - (-10))\,\frac{10\,\text{k}}{10\,\text{k} + 10\,\text{k}} = 0\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 5\,\text{k}\Omega$$

5. C_2 is a capacitor $\Rightarrow$ open at DC. No DC current can cross it, so everything on its left is disconnected: start fresh after it.

6. Finally, collect everything at the output node V_o.
7. First DC path here: R_3 to the $+10\,\mathrm{V}$ supply, so $(V_{th}, R_{th}) = (10\,\mathrm{V},\ 20\,\mathrm{k\Omega})$.
8. Add R_4 to ground. Divider between $V_a = 10\,\mathrm{V}$ (through $R_a = 20\,\mathrm{k\Omega}$) and $V_b = 0\,\mathrm{V}$ (through $R_b = 10\,\mathrm{k\Omega}$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 0 + (10 - (0))\,\frac{10\,\mathrm{k}}{20\,\mathrm{k} + 10\,\mathrm{k}} = 3.33\,\mathrm{V}, \qquad R_{th} = R_a \parallel R_b = 6.67\,\mathrm{k\Omega}$$

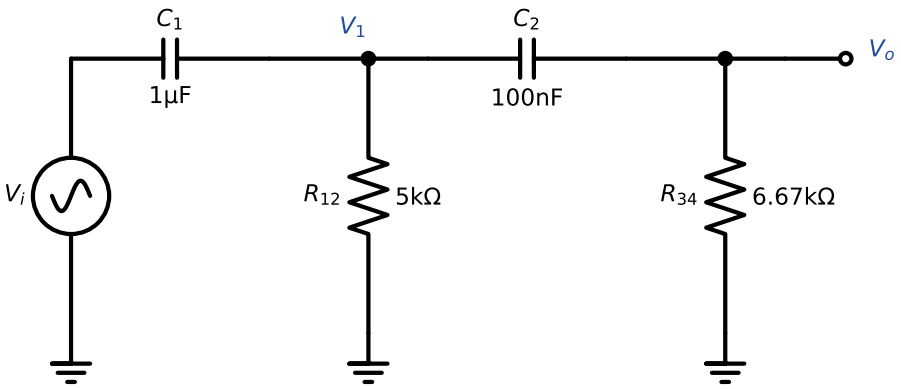
$$\boxed{V_{o,dc} = 3.33\,\mathrm{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1, R_2 both go from the same node to (AC) ground, so they are in parallel: $R_{12} = R_1 \parallel R_2 = 10\,\mathrm{k} \parallel 10\,\mathrm{k} = 5\,\mathrm{k\Omega}$.
- R_3, R_4 both go from the same node to (AC) ground, so they are in parallel: $R_{34} = R_3 \parallel R_4 = 20\,\mathrm{k} \parallel 10\,\mathrm{k} = 6.67\,\mathrm{k\Omega}$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_{12} \parallel \left(\frac{1}{sC_2} + R_{34}\right) = \frac{2860\,s + 4.29 \times 10^6}{s + 857}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^2 + 1500\,s}{s^2 + 1850\,s + 3 \times 10^5}$$
$$\frac{V_o}{V_1} = \frac{R_{34}}{\frac{1}{sC_2} + R_{34}} = \frac{s}{s + 1500}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{s^2}{s^2 + 1850\,s + 3 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s^2 \Rightarrow$ 2 zeros at $s = 0$.
- Denominator: $s^2 + 1850\,s + 3 \times 10^5$; quadratic formula:

$$s = \frac{-1850 \pm \sqrt{(1850)^2 - 4(3 \times 10^5)}}{2} = \frac{-1850 \pm 1490}{2} \Rightarrow s = -180,\ -1670$$

Factored form (leading coefficient in front):

$$H(s) = 1\,\frac{s^2}{(s + 180)(s + 1670)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1\,\frac{s}{s + 2\pi \cdot 28.6}\,\frac{s}{s + 2\pi \cdot 266}, \qquad K = 1$$

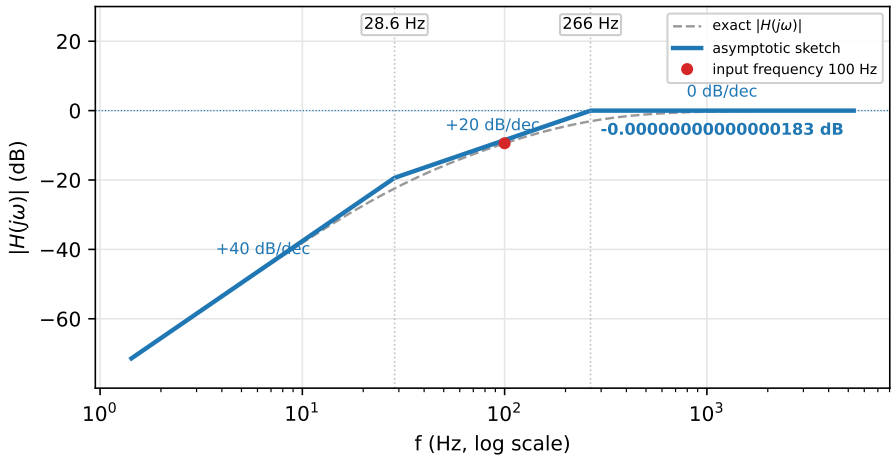
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	180	28.6 Hz	−20 dB/dec
pole	1670	266 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 3.33 \times 10^{-6}\,s^2$, so the plot starts with a slope of +40 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 28.6 Hz	+40 dB/dec	-19.4 dB at end
28.6 Hz – 266 Hz	+20 dB/dec	-19.4 dB at start, 0 dB at end
$f >$ 266 Hz	0 dB/dec (flat)	0 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}1 = 0\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.3\sin(2\pi\cdot 100\,t)$, so $\omega_{in} = 2\pi\cdot 100 = 628\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s+2\pi\cdot 28.6}$	0.9615	0.278
$\dfrac{s}{s+2\pi\cdot 266}$	0.3521	1.21
total	**0.3385** (product)	**1.49** (sum)

$$|H(j\omega_{in})| = 0.339, \qquad \angle H(j\omega_{in}) = 1.49 \text{ rad}$$
$$v_{o,ac}(t) = 0.339\times 0.3\,\sin(2\pi\cdot 100\,t + 1.49) = 0.102\,\sin(2\pi\cdot 100\,t + 1.49)$$

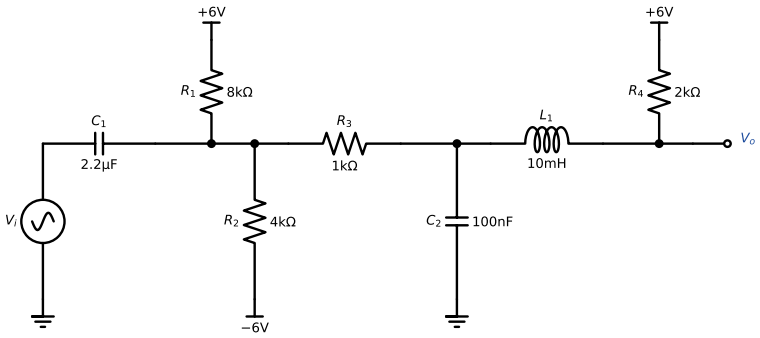
Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = 3.33 + 0.102\,\sin(2\pi\cdot 100\,t + 1.49)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{s}{s+2\pi\cdot 28.6}\,\frac{s}{s+2\pi\cdot 266} \quad \text{(corners: 28.6 Hz, 266 Hz)}$$
$$V_o(t) = 3.33 + 0.102\,\sin(2\pi\cdot 100\,t + 1.49)\ \text{V}$$

Solution 1B.16 **Hard**

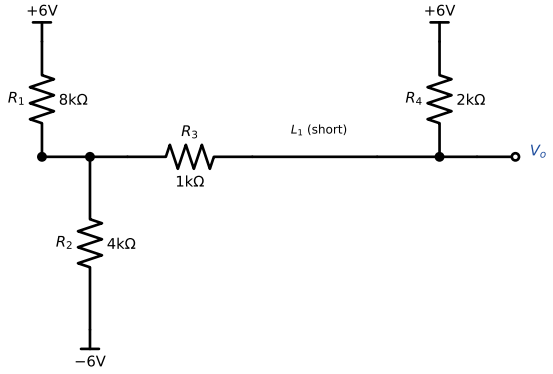


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies $-6\,\text{V}$, $+6\,\text{V}$. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1, C_2); every inductor is a short circuit (L_1). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal $0\,\text{V}$ source: $(V_{th}, R_{th}) = (0\,\text{V},\, 0\,\Omega)$.
2. C_1 is a capacitor $\Rightarrow$ open at DC. No DC current can cross it, so everything on its left is disconnected: start fresh after it.
3. First DC path here: R_1 to the $+6\,\text{V}$ supply, so $(V_{th}, R_{th}) = (6\,\text{V},\ 8\,\text{k}\Omega)$.
4. Add R_2 to the $-6\,\text{V}$ supply. Divider between $V_a = 6\,\text{V}$ (through $R_a = 8\,\text{k}\Omega$) and $V_b = -6\,\text{V}$ (through $R_b = 4\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = -6 + (6 - (-6))\,\frac{4\,\text{k}}{8\,\text{k} + 4\,\text{k}} = -2\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 2.67\,\text{k}\Omega$$

5. Walk through R_3 ($1\,\text{k}\Omega$ at DC). A series resistor just adds to R_{th}: $(V_{th}, R_{th}) = (-2\,\text{V},\ 3.67\,\text{k}\Omega)$.

6. C_2 (at V_2) is a capacitor: open at DC, ignore it.
7. L_1 is a short at DC (inductor), so the pair is unchanged.
8. Finally, collect everything at the output node V_o.
9. Add R_4 to the $+6\,\mathrm{V}$ supply. Divider between $V_a = -2\,\mathrm{V}$ (through $R_a = 3.67\,\mathrm{k\Omega}$) and $V_b = 6\,\mathrm{V}$ (through $R_b = 2\,\mathrm{k\Omega}$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 6 + (-2 - (6))\,\frac{2\,\mathrm{k}}{3.67\,\mathrm{k} + 2\,\mathrm{k}} = 3.18\,\mathrm{V}, \qquad R_{th} = R_a \parallel R_b = 1.29\,\mathrm{k\Omega}$$

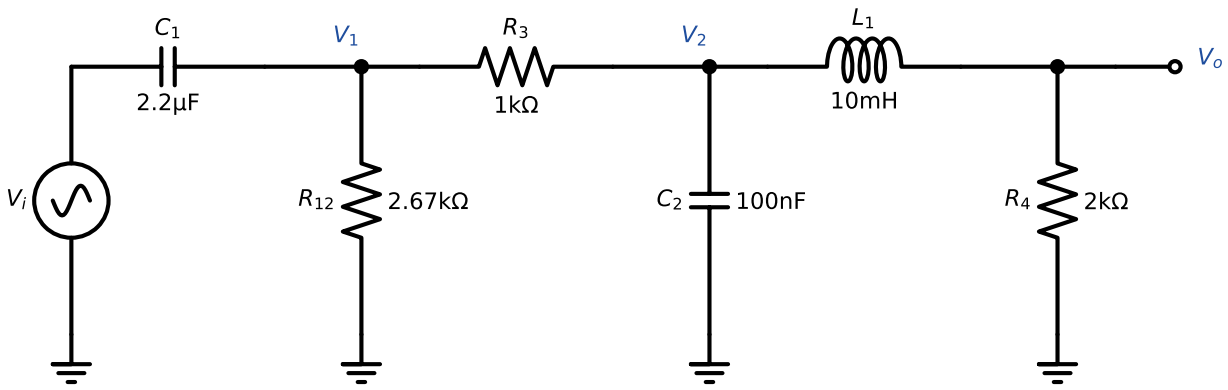
$$\boxed{V_{o,dc} = 3.18\,\mathrm{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1, R_2 both go from the same node to (AC) ground, so they are in parallel: $R_{12} = R_1 \parallel R_2 = 8\,\mathrm{k} \parallel 4\,\mathrm{k} = 2.67\,\mathrm{k\Omega}$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_2 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_2}{V_1} \cdot \frac{V_o}{V_2}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_2 = \frac{1}{sC_2} \parallel (sL_1 + R_4) = \frac{10^7\, s + 2 \times 10^{12}}{s^2 + 2 \times 10^5\, s + 10^9}$$

$$Z_1 = R_{12} \parallel (R_3 + Z_2) = \frac{727\, s^2 + 1.53 \times 10^8\, s + 2.18 \times 10^{12}}{s^2 + 2.03 \times 10^5\, s + 1.55 \times 10^9}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^3 + 2.1 \times 10^5\, s^2 + 3 \times 10^9\, s}{s^3 + 2.11 \times 10^5\, s^2 + 3.13 \times 10^9\, s + 9.66 \times 10^{11}}$$
$$\frac{V_2}{V_1} = \frac{Z_2}{R_3 + Z_2} = \frac{10000\, s + 2 \times 10^9}{s^2 + 2.1 \times 10^5\, s + 3 \times 10^9}$$
$$\frac{V_o}{V_2} = \frac{R_4}{sL_1 + R_4} = \frac{2 \times 10^5}{s + 2 \times 10^5}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{2 \times 10^9\, s}{s^3 + 2.11 \times 10^5\, s^2 + 3.13 \times 10^9\, s + 9.66 \times 10^{11}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^3 + 2.11 \times 10^5\, s^2 + 3.13 \times 10^9\, s + 9.66 \times 10^{11}$; this is degree 3: use a calculator's polynomial solver (or `numpy.roots`) $\Rightarrow s = -316,\ -15700,\ -1.95 \times 10^5$.

Factored form (leading coefficient in front):

$$H(s) = 2 \times 10^9\, \frac{s}{(s + 316)(s + 15700)(s + 1.95 \times 10^5)}$$

Bode-ready form: rewrite every $(s + p) = p\left(1 + \frac{s}{p}\right)$ and $(s + z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.654\, \frac{s}{s + 2\pi \cdot 50.2}\, \frac{1}{1 + \frac{s}{2\pi \cdot 2.5\text{k}}}\, \frac{1}{1 + \frac{s}{2\pi \cdot 31\text{k}}}, \qquad K = 0.654$$

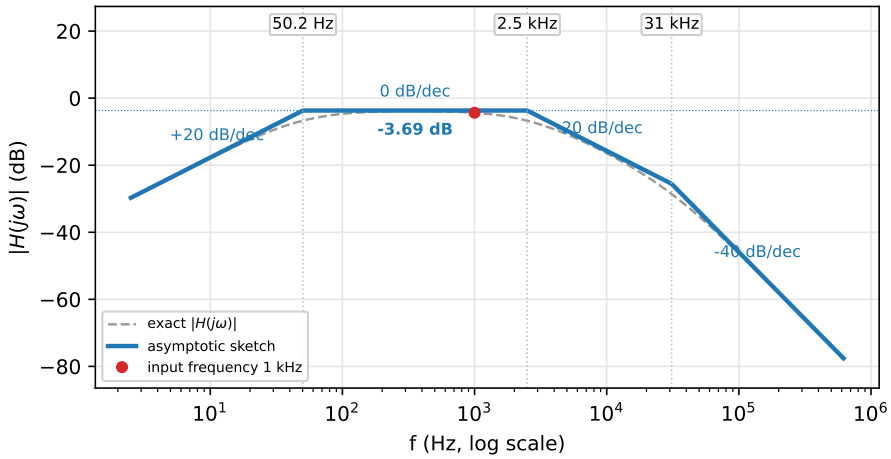
type	ω_c (rad/s)	$f_c = \omega_c / 2\pi$	slope change
pole	316	50.2 Hz	−20 dB/dec
pole	15700	2.5 kHz	−20 dB/dec
pole	1.95×10^5	31 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.00207\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 50.2$ Hz	+20 dB/dec	-3.69 dB at end
50.2 Hz – 2.5 kHz	0 dB/dec (flat)	-3.69 dB at start, -3.69 dB at end
2.5 kHz – 31 kHz	−20 dB/dec	-3.69 dB at start, -25.5 dB at end
$f > 31$ kHz	−40 dB/dec	-25.5 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}0.654 = -3.69\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\cos(2\pi\cdot 1000\,t)$, so $\omega_{in} = 2\pi\cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.654	0.6535	0
$\dfrac{s}{s+2\pi\cdot 50.2}$	0.9987	0.0502
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 2.5\text{k}}}$	0.9286	−0.38
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 31\text{k}}}$	0.9995	−0.0323
total	**0.6058** (product)	**−0.362** (sum)

$$|H(j\omega_{in})| = 0.606, \qquad \angle H(j\omega_{in}) = -0.362 \text{ rad}$$

$$v_{o,ac}(t) = 0.606 \times 1 \cos\left(2\pi \cdot 1000\, t - 0.362\right) = 0.606 \cos\left(2\pi \cdot 1000\, t - 0.362\right)$$

Add the DC part from Step 2 (superposition):

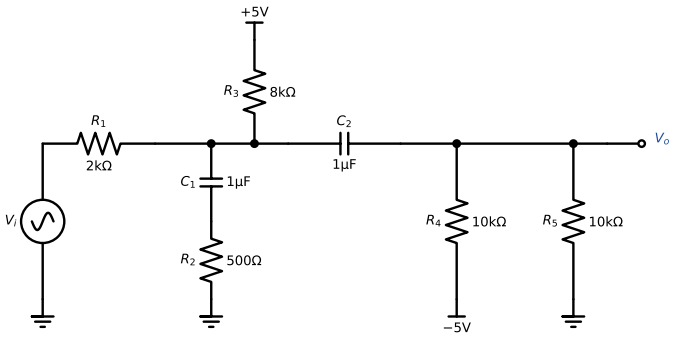
$$\boxed{V_o(t) = 3.18 + 0.606 \cos\left(2\pi \cdot 1000\, t - 0.362\right) \text{ V}}$$

Answer

$$H(s) = 0.654 \frac{s}{s + 2\pi \cdot 50.2} \frac{1}{1 + \dfrac{s}{2\pi \cdot 2.5\text{k}}} \frac{1}{1 + \dfrac{s}{2\pi \cdot 31\text{k}}} \quad \text{(corners: 50.2 Hz, 2.5 kHz, 31 kHz)}$$

$$V_o(t) = 3.18 + 0.606 \cos\left(2\pi \cdot 1000\, t - 0.362\right) \text{ V}$$

Solution 1B.17 **Hard**

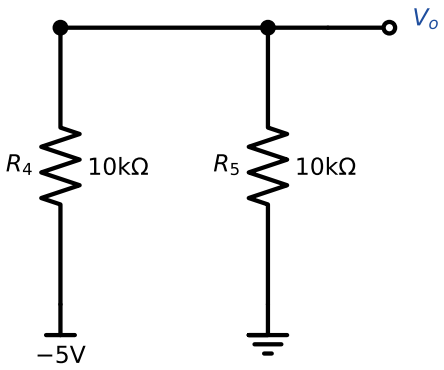


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies $-5\,\text{V}$, $+5\,\text{V}$. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1, C_2). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal $0\,\text{V}$ source: $(V_{th}, R_{th}) = (0\,\text{V},\, 0\,\Omega)$.
2. Walk through R_1 ($2\,\text{k}\Omega$ at DC). A series resistor just adds to R_{th}: $(V_{th}, R_{th}) = (0\,\text{V},\ 2\,\text{k}\Omega)$.
3. $C_1 + R_2$ (at V_1) contains a capacitor: open at DC, ignore it.
4. Add R_3 to the $+5\,\text{V}$ supply. Divider between $V_a = 0\,\text{V}$ (through $R_a = 2\,\text{k}\Omega$) and $V_b = 5\,\text{V}$ (through $R_b = 8\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 5 + (0 - (5))\,\frac{8\,\text{k}}{2\,\text{k} + 8\,\text{k}} = 1\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 1.6\,\text{k}\Omega$$

5. C_2 is a capacitor $\Rightarrow$ open at DC. No DC current can cross it, so everything on its left is disconnected: start fresh after it.

6. Finally, collect everything at the output node V_o.
7. First DC path here: R_4 to the $-5\,\mathrm{V}$ supply, so $(V_{th}, R_{th}) = (-5\,\mathrm{V},\ 10\,\mathrm{k\Omega})$.
8. Add R_5 to ground. Divider between $V_a = -5\,\mathrm{V}$ (through $R_a = 10\,\mathrm{k\Omega}$) and $V_b = 0\,\mathrm{V}$ (through $R_b = 10\,\mathrm{k\Omega}$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 0 + (-5 - (0))\,\frac{10\,\mathrm{k}}{10\,\mathrm{k} + 10\,\mathrm{k}} = -2.5\,\mathrm{V}, \qquad R_{th} = R_a \parallel R_b = 5\,\mathrm{k\Omega}$$

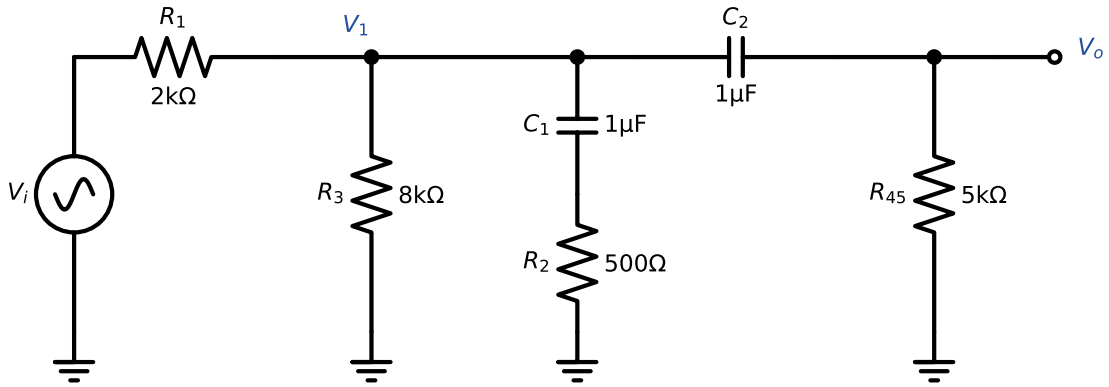
$$\boxed{V_{o,dc} = -2.5\,\mathrm{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_4, R_5 both go from the same node to (AC) ground, so they are in parallel: $R_{45} = R_4 \parallel R_5 = 10\,\mathrm{k} \parallel 10\,\mathrm{k} = 5\,\mathrm{k\Omega}$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_3 \parallel \left(\frac{1}{sC_1} + R_2\right) \parallel \left(\frac{1}{sC_2} + R_{45}\right) = \frac{430\,s^2 + 9.46 \times 10^5\,s + 1.72 \times 10^8}{s^2 + 462\,s + 21500}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{R_1 + Z_1} = \frac{0.177\,s^2 + 389\,s + 70800}{s^2 + 770\,s + 88500}$$

$$\frac{V_o}{V_1} = \frac{R_{45}}{\frac{1}{sC_2} + R_{45}} = \frac{s}{s + 200}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{0.177\,s^2 + 354\,s}{s^2 + 770\,s + 88500}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Numerator (after taking out the s factor): $0.177\,s + 354$; divide by the leading coefficient 0.177 (this does not move the roots): $s + 2000$; $s + 2000 = 0 \Rightarrow s = -2000$.
- Denominator: $s^2 + 770\,s + 88500$; quadratic formula:

$$s = \frac{-770 \pm \sqrt{(770)^2 - 4(88500)}}{2} = \frac{-770 \pm 489}{2} \Rightarrow s = -141,\ -629$$

Factored form (leading coefficient in front):

$$H(s) = 0.177\,\frac{s(s + 2000)}{(s + 141)(s + 629)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

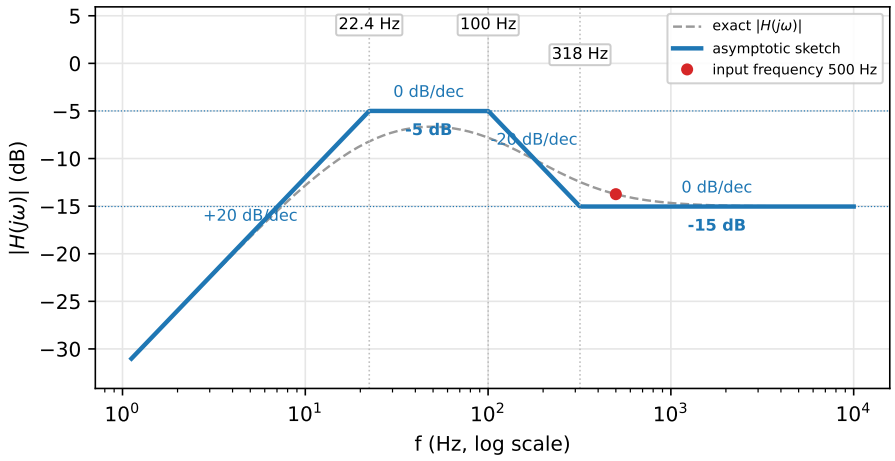
$$H(s) = 0.563\,\frac{s}{s + 2\pi \cdot 22.4}\,\frac{1}{1 + \frac{s}{2\pi \cdot 100}}\left(1 + \frac{s}{2\pi \cdot 318}\right), \qquad K = 0.563$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	141	22.4 Hz	−20 dB/dec
pole	629	100 Hz	−20 dB/dec
zero	2000	318 Hz	+20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.004\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 22.4$ Hz	+20 dB/dec	-5 dB at end
22.4 Hz – 100 Hz	0 dB/dec (flat)	-5 dB at start, -5 dB at end
100 Hz – 318 Hz	−20 dB/dec	-5 dB at start, -15 dB at end
$f > 318$ Hz	0 dB/dec (flat)	-15 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\cos\left(2\pi\cdot 500\,t - \frac{\pi}{3}\right)$, so $\omega_{in} = 2\pi\cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.563	0.5625	0
$\dfrac{s}{s+2\pi\cdot 22.4}$	0.999	0.0447
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100}}$	0.1964	-1.37
$\left(1+\dfrac{s}{2\pi\cdot 318}\right)$	1.862	1
total	**0.2055** (product)	**−0.324** (sum)

$$|H(j\omega_{in})| = 0.206, \qquad \angle H(j\omega_{in}) = -0.324 \text{ rad}$$
$$v_{o,ac}(t) = 0.206\times 1\cos\left(2\pi\cdot 500\,t - 1.37\right) = 0.206\cos\left(2\pi\cdot 500\,t - 1.37\right)$$

(phase: $\theta + \angle H = -\frac{\pi}{3} + (-0.324) = -1.37\,\text{rad}$.) Add the DC part from Step 2 (superposition):

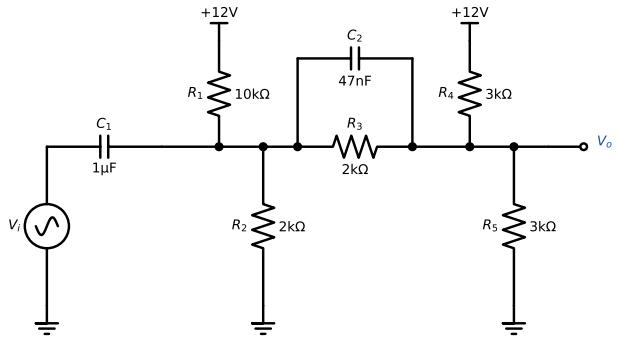
$$\boxed{V_o(t) = -2.5 + 0.206\cos\left(2\pi\cdot 500\,t - 1.37\right)\ \text{V}}$$

Answer

$$H(s) = 0.563\,\frac{s}{s + 2\pi \cdot 22.4}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 100}}\left(1 + \frac{s}{2\pi \cdot 318}\right) \quad \text{(corners: 22.4 Hz, 100 Hz, 318 Hz)}$$

$$V_o(t) = -2.5 + 0.206\,\cos\left(2\pi \cdot 500\,t - 1.37\right)\ \text{V}$$

Solution 1B.18 **Hard**

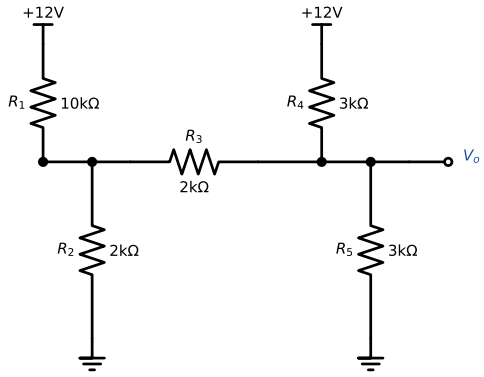


Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supplies +12 V. Because there are both DC and AC sources, use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. Solve the DC circuit (Step 2) and the AC circuit (Steps 3–7) separately, then add.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: the AC input is set to zero ($V_i = 0$: a wire to ground); every capacitor is an open circuit (C_1, C_2). Only the part of the circuit that is still connected to V_o matters:



Reduce the circuit with one tool only — the *divider between two voltages* (tool T3) — walking toward V_o and keeping a Thevenin pair (V_{th}, R_{th}) for everything already visited.

1. Start at the input node: since $V_i = 0$ it is tied to ground, i.e. it is an ideal 0 V source: $(V_{th}, R_{th}) = (0\,\text{V},\, 0\,\Omega)$.
2. C_1 is a capacitor $\Rightarrow$ open at DC. No DC current can cross it, so everything on its left is disconnected: start fresh after it.
3. First DC path here: R_1 to the +12 V supply, so $(V_{th}, R_{th}) = (12\,\text{V},\ 10\,\text{k}\Omega)$.
4. Add R_2 to ground. Divider between $V_a = 12$ V (through $R_a = 10\,\text{k}\Omega$) and $V_b = 0$ V (through $R_b = 2\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 0 + (12 - (0))\,\frac{2\,\text{k}}{10\,\text{k} + 2\,\text{k}} = 2\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 1.67\,\text{k}\Omega$$

5. Walk through $R_3 \parallel C_2$ (2 kΩ at DC). A series resistor just adds to R_{th}: $(V_{th}, R_{th}) = (2\,\text{V},\ 3.67\,\text{k}\Omega)$.

6. Finally, collect everything at the output node V_o.

7. Add R_4 to the $+12\,\text{V}$ supply. Divider between $V_a = 2\,\text{V}$ (through $R_a = 3.67\,\text{k}\Omega$) and $V_b = 12\,\text{V}$ (through $R_b = 3\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 12 + (2 - (12))\,\frac{3\,\text{k}}{3.67\,\text{k} + 3\,\text{k}} = 7.5\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 1.65\,\text{k}\Omega$$

8. Add R_5 to ground. Divider between $V_a = 7.5\,\text{V}$ (through $R_a = 1.65\,\text{k}\Omega$) and $V_b = 0\,\text{V}$ (through $R_b = 3\,\text{k}\Omega$):

$$V_{th} = V_b + (V_a - V_b)\frac{R_b}{R_a + R_b} = 0 + (7.5 - (0))\,\frac{3\,\text{k}}{1.65\,\text{k} + 3\,\text{k}} = 4.84\,\text{V}, \qquad R_{th} = R_a \parallel R_b = 1.06\,\text{k}\Omega$$

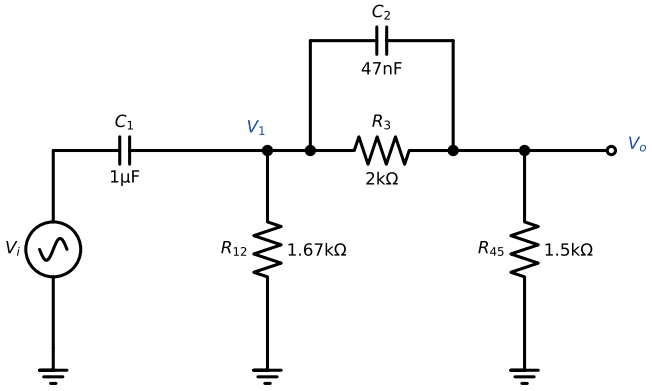
$$\boxed{V_{o,dc} = 4.84\,\text{V}}$$

Step 3 — AC circuit

Set every DC supply to zero: each supply rail becomes AC ground. Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1, R_2 both go from the same node to (AC) ground, so they are in parallel: $R_{12} = R_1 \parallel R_2 = 10\,\text{k} \parallel 2\,\text{k} = 1.67\,\text{k}\Omega$.
- R_4, R_5 both go from the same node to (AC) ground, so they are in parallel: $R_{45} = R_4 \parallel R_5 = 3\,\text{k} \parallel 3\,\text{k} = 1.5\,\text{k}\Omega$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_{12} \parallel \left(\left(R_3 \parallel \frac{1}{sC_2}\right) + R_{45}\right) = \frac{789\,s + 1.96 \times 10^7}{s + 17400}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^2 + 24800\,s}{s^2 + 26100\,s + 2.2 \times 10^7}$$
$$\frac{V_o}{V_1} = \frac{R_{45}}{R_3 \parallel \frac{1}{sC_2} + R_{45}} = \frac{s + 10600}{s + 24800}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{s^2 + 10600\,s}{s^2 + 26100\,s + 2.2 \times 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Numerator (after taking out the s factor): $s + 10600$; $s + 10600 = 0 \Rightarrow s = -10600$.
- Denominator: $s^2 + 26100\,s + 2.2 \times 10^7$; quadratic formula:

$$s = \frac{-26100 \pm \sqrt{(26100)^2 - 4(2.2 \times 10^7)}}{2} = \frac{-26100 \pm 24300}{2} \Rightarrow s = -872,\ -25200$$

Factored form (leading coefficient in front):

$$H(s) = 1\,\frac{s(s + 10600)}{(s + 872)(s + 25200)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

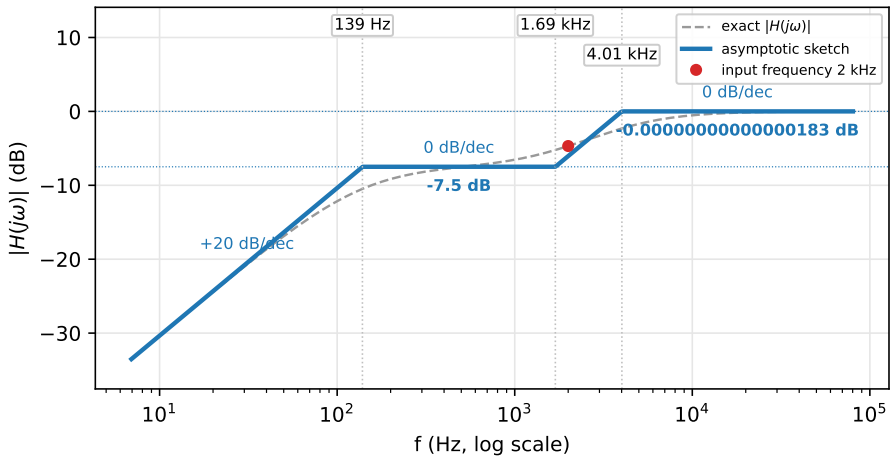
$$H(s) = 0.422\,\frac{s}{s + 2\pi \cdot 139}\left(1 + \frac{s}{2\pi \cdot 1.69\text{k}}\right)\frac{1}{1 + \frac{s}{2\pi \cdot 4.01\text{k}}}, \qquad K = 0.422$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	872	139 Hz	−20 dB/dec
zero	10600	1.69 kHz	+20 dB/dec
pole	25200	4.01 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 4.84 \times 10^{-4}\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 139$ Hz	+20 dB/dec	-7.5 dB at end
139 Hz – 1.69 kHz	0 dB/dec (flat)	-7.5 dB at start, -7.5 dB at end
1.69 kHz – 4.01 kHz	+20 dB/dec	-7.5 dB at start, 0 dB at end
$f > 4.01$ kHz	0 dB/dec (flat)	0 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5 \sin\left(2\pi \cdot 2000\, t + \frac{\pi}{6}\right)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.422	0.4219	0
$\dfrac{s}{s + 2\pi \cdot 139}$	0.9976	0.0693
$\left(1 + \dfrac{s}{2\pi \cdot 1.69\text{k}}\right)$	1.548	0.868
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 4.01\text{k}}}$	0.895	−0.462
total	**0.583** (product)	**0.475** (sum)

$$|H(j\omega_{in})| = 0.583, \qquad \angle H(j\omega_{in}) = 0.475 \text{ rad}$$
$$v_{o,ac}(t) = 0.583 \times 0.5 \sin\left(2\pi \cdot 2000\, t + 0.999\right) = 0.291 \sin\left(2\pi \cdot 2000\, t + 0.999\right)$$

(phase: $\theta + \angle H = \frac{\pi}{6} + (0.475) = 0.999\,\text{rad.}$) Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = 4.84 + 0.291\,\sin\left(2\pi \cdot 2000\,t + 0.999\right)\ \text{V}}$$

Answer

$$H(s) = 0.422\,\frac{s}{s + 2\pi \cdot 139}\left(1 + \frac{s}{2\pi \cdot 1.69\text{k}}\right)\frac{1}{1 + \dfrac{s}{2\pi \cdot 4.01\text{k}}} \quad \text{(corners: 139 Hz, 1.69 kHz, 4.01 kHz)}$$

$$V_o(t) = 4.84 + 0.291\,\sin\left(2\pi \cdot 2000\,t + 0.999\right)\ \text{V}$$